BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE-306
POWER SYSTEM I LABORATORY

Final Project Report
Section:C1 Group: 3

REACTIVE POWER COMPENSATION IN GRID CONNECTED PV CELL

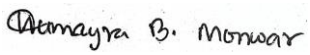
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ABSTRACT

This project investigates the application of reactive power compensation techniques in grid-connected photovoltaic (PV) systems using Static Synchronous Compensators (STATCOMs) and fixed capacitor banks to reduce active and reactive power demand from the grid by supplying variable reactive power from an alternative source, thereby decreasing reliance on utility power and facilitating renewable energy integration. The study presents comprehensive case studies comparing the application of fixed capacitor banks and STATCOMs for dynamic VAR compensation in a grid-connected PV system. The behavior of dynamic loads, such as induction motors, under varying steady-state operation conditions and reactive power demands is analyzed. System modeling and analysis are conducted for both scenarios, considering various load demands and network conditions. The system integrates various load dynamics and changing solar irradiance conditions are simulated to assess the system's performance and response to disturbances. This project proposes a design methodology for reactive compensation in PV-integrated grid systems employing both STATCOMs and fixed capacitor banks. Detailed simulations and comparisons between the two compensation techniques are provided, highlighting the advantages and limitations of each approach under different operating conditions.

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1 INTRODUCTION:

As the demand for electricity increases with the world's growing infrastructure and population, it is crucial to discover methods of generating power that don't damage the environment.. Traditional power plants, like those fueled by coal or gas, cause pollution and need a lot of space and resources. To tackle these problems, renewable energy sources like solar power are becoming increasingly popular.Solar power has many benefits. It's easy to set up, doesn't take up much space, and the cost of making solar panels is going down. But to make solar energy work well, we need to connect the solar panels in the right way. By connecting them in a series or parallel pattern, we can make sure they produce enough voltage and current to power our devices.When we connect solar panels to the main electricity grid, we have to do it carefully. There are different ways to connect them, like using a single stage or double stage connection. Single-stage connections are simple, but they may not work well if the voltage from the solar panels isn't right. Double-stage connections are more flexible because they can adjust to different voltage levels.One big challenge with solar power is managing something called reactive power. Reactive power helps keep the voltage stable in electrical systems. Traditional solutions to manage reactive power, like using fixed capacitor banks, have limitations. These banks can't change how much power they provide based on the needs of the system, which can cause problems like inefficient power use and extra costs.To tackle these issues, we're looking at using something called a STATCOM in solar power systems. A STATCOM can adjust how much reactive power it provides in real-time, helping keep the system running smoothly. This project explores how STATCOMs could improve solar power systems by dynamically managing reactive power . Traditionally, STATCOMs have been employed in transmission networks for voltage regulation and VAR compensation.this project explores their integration into distribution networks, specifically in PV grid-connected systems utilizing distributed energy resources Throughout this project, we'll explore different aspects of solar power systems, from how solar panels are designed to how we connect them to the grid. We'll also dive into the details of using STATCOMs and fixed capacitor banks for managing reactive power. By running simulations in different situations, hoping to learn more about how these systems can work together to make solar power even more efficient and reliable.

2 DESIGN:

2.1 Problem formulation

2.1.1 Identification of scope:

The scope of this project encompasses investigating reactive power compensation techniques in grid-connected PV systems using STATCOMs and fixed capacitor banks. It involves studying their integration into distribution networks to reduce grid dependency and promote renewable energy integration. The project includes comprehensive case studies comparing fixed capacitor banks and STATCOMs for dynamic VAR compensation, analyzing their performance under varying load demands and network conditions. System modeling and simulations on a 100-kW rated PV grid-connected system using MATLAB/Simulink form an integral part of the scope, aiming to develop a design methodology that addresses the challenges and opportunities associated with reactive compensation in PV-integrated grid systems.

2.1.2 Literature Review:

Reactive power compensation in grid-connected photovoltaic (PV) systems has been a subject of significant research interest in recent years. The concept of Reactive Power Compensation embraces a wide and diverse field of both system and customer problems, especially related with power quality issues, since most of power quality problems can be attenuated or solved with an adequate control of reactive power,like-

- The need to maintain the stability of synchronous machines
- The need to regulate voltage profiles in the network to prevent unnecessary flows of reactive power on transmission lines
- The need to control voltage within acceptable bounds about the desired steady-state value to provide quality service to consumer loads

With the increasing deployment of renewable energy sources like solar power, the need to manage reactive power becomes crucial for grid stability and efficiency. STATCOMs have traditionally been utilized in transmission networks for voltage regulation and VAR compensation. However, recent studies have explored their potential integration into distribution networks, particularly in the context of PV grid-connected systems..Fixed capacitor banks have also been widely used for reactive power compensation in various electrical systems. However, their static nature and inability to adapt to varying load conditions limit their effectiveness, especially in dynamic PV systems . Several studies have compared the performance of fixed capacitor banks and STATCOMs in PV systems, highlighting the advantages of STATCOMs in providing flexible and dynamic reactive power support .Moreover, the integration of STATCOMs with PV systems offers additional benefits such as improved grid integration and enhanced renewable energy penetration. By supplying variable reactive power from an

alternative source, STATCOMs reduce dependency on the utility grid and promote renewable energy integration.In terms of simulation and analysis, MATLAB/Simulink has emerged as a widely used tool for modeling and simulating PV grid-connected systems with reactive power compensation techniques. The flexibility and versatility of MATLAB/Simulink enable researchers to simulate various load dynamics, network conditions, and control strategies, facilitating comprehensive analysis and optimization of PV systems with reactive power compensation.Overall, the literature highlights the importance of effective reactive power compensation in PV grid-connected systems and the potential of STATCOMs to address this challenge. By leveraging advanced simulation tools and methodologies, researchers aim to develop optimized designs and control strategies for enhancing the performance and stability of PV systems in the context of renewable energy integration.

2.1.3 Formulation of Problem:

The problem this project addresses is how to effectively compensate for reactive power in grid-connected photovoltaic (PV) systems using Static Synchronous Compensators (STATCOMs) and fixed capacitor banks. With traditional use in transmission networks, STATCOMs are now being explored for integration into distribution networks to support PV grid-connected systems. The main goal is to decrease reactive power demand from the grid by providing variable reactive power from another source, thereby encouraging the integration of renewable energy. By comparing fixed capacitor banks and STATCOMs for dynamic VAR compensation, this study aims to understand their effectiveness in managing varying loads, such as induction motors, under different conditions. Through system modeling and simulations using MATLAB/Simulink, the project seeks to develop a design methodology that considers different load demands and network conditions, highlighting the strengths and limitations of each approach in optimizing PV system performance.The simulation will be based on a rated PV grid-connected system, utilizing MATLAB/Simulink for modeling and analysis.

2.2 Design Method:

The design method for the solar photovoltaic (PV) grid system with STATCOM and fixed capacitor involves several steps to ensure the efficient and reliable operation of the system components. These steps include the design of the solar PV array system, DC-DC boost converter, grid-tied inverter LCL harmonic filter, LCL harmonic filter for STATCOM ,fixed capacitor bank ratings and STATCOM.

2.2.1 Design of Solar PV Array System:

The solar PV panel system is created through parallel and series integration of different solar PV panels. Series connection increases the solar PV panel array terminal voltage where as parallel connection increases the solar PV panel array current rating.

Specifications of a single panel:

- Rating: 213.15 W
- Maximum Power Point (MPP) Voltage: 29 V
- Open Circuit Voltage: 36.3 V
- Short Circuit Current: 7.84 A
- Maximum Power Point Current: 7.35

For the solar PV array:

- Series-connected PV panels: 10
- Parallel-connected PV panels: 47
- Overall rating: 100.345 kW
- PV panel voltage at open circuit: 363 V
- PV current at short circuit: 368.48 A
- Maximum power point current: 345.45 A
- Maximum power point voltage: 290 V

Power-voltage and current-voltage characteristics of the solar PV array for different irradiance conditions are depicted in figure

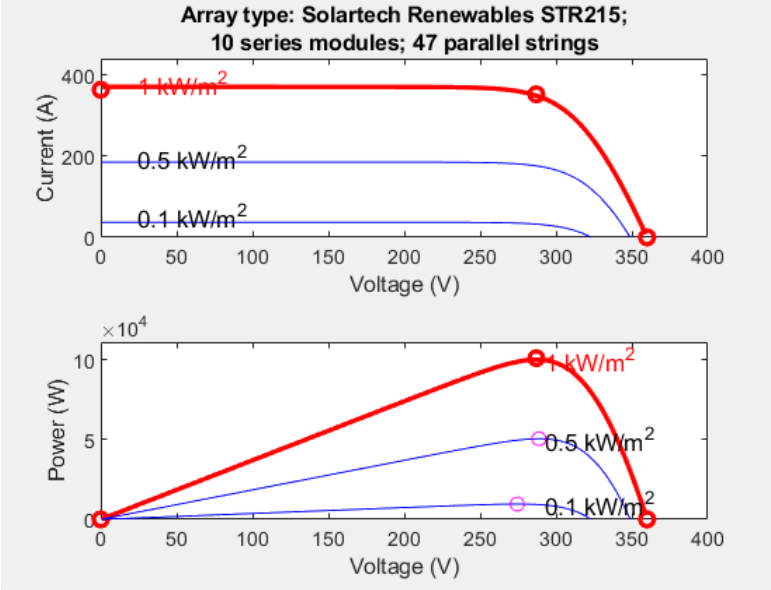


Figure 1:Fig: Power-voltage and current-voltage characteristics of the solar PV array for different irradiance conditions

.

The solar PV array exhibits varying maximum power outputs:

- 9.72 kW at 100 W/m² and 25°C
- 50.75 kW at 500 W/m² and 25°C
- 100.345 kW at 1000 W/m² and 25°C.

2.2.2 Design of DC-DC Boost Converter:

The direct current – direct current step-up converter is designed to meet the voltage and power requirements of the solar PV array and the output voltage demand of the grid-tied inverter.

Components :

- Boost Converter Components:
- Input voltage source
- Switch (typically an IGBT)
- Inductor
- Diode
- Capacitor
- Load resistor

Specifications:

Solar PV array ratings: 290 V, 100.345 kW (standard testing conditions).

Grid-tied inverter output voltage requirement: 700 V.

$$I_{outmax} = \frac{P}{V_{out}}$$

$$\Delta I_L = 0.07 \times I_{outmax} \times \frac{V_{out}}{V_{in}}$$

$$\Delta V_{out} = 0.007 \times V_{out}$$

$$L = \frac{V_{in} \times (V_{out} - V_{in})}{\Delta I_L \times F_s \times V_{out}}$$

$$C = \frac{I_{outmax} \times (1 - \frac{V_{in}}{V_{out}})}{\Delta V_{out} \times F_s}$$

The maximum current of the direct current – direct current step-up converter at the output terminal is,

$$I_{outmax} = \frac{P}{V_{out}} = \frac{100.345 \times 1000}{700} = 143.35\ A$$

The ripple inductor current of the direct current – direct current step-up converter is,

$$\Delta I_L = 0.07 \times I_{outmax} \times \frac{V_{out}}{V_{in}} = 0.07 \times 143.34 \times \frac{700}{290} = 24.21\ A$$

The ripple inductor voltage ,

$$\Delta V_{out} = 0.007 \times V_{out} = 0.007 \times 700 = 4.9\text{V}$$

The designed value of the inductor,

$$L = \frac{V_{in} \times (V_{out} - V_{in})}{\Delta I_L \times F_S \times V_{out}} = \frac{290 \times (700 - 290)}{24.21 \times 5000 \times 700} = 1.4\ mH$$

The designed value of the capacitor

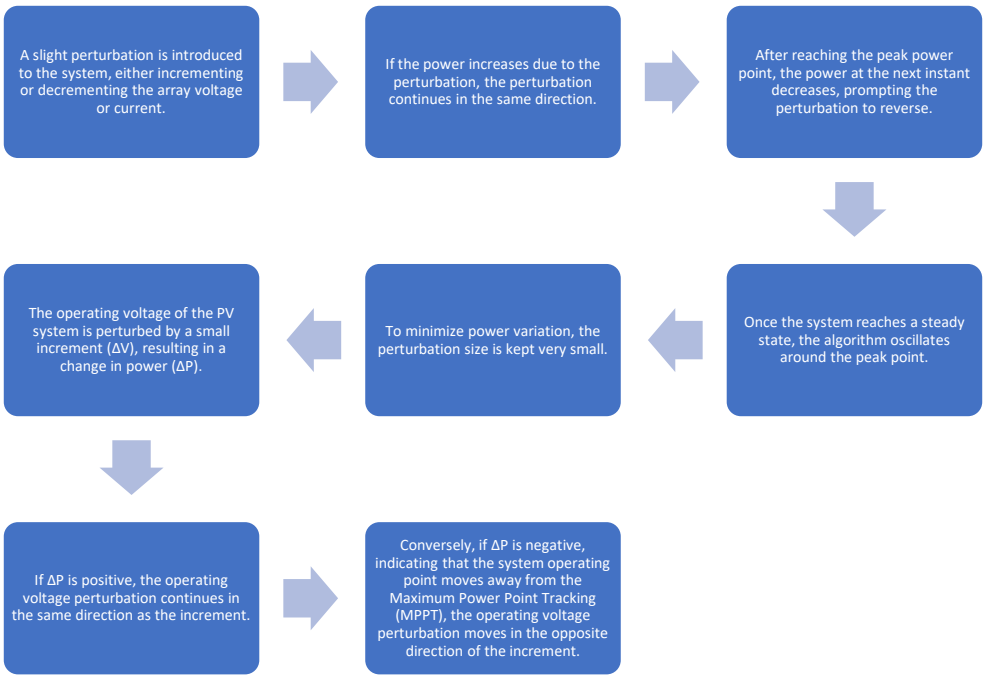
$$C = \frac{I_{outmax} \times (1 - \frac{V_{in}}{V_{out}})}{\Delta V_{out} \times F_S} = \frac{143.35 \times (1 - \frac{290}{700})}{4.9 \times 5000} = 3400\ \mu F$$

The switch, typically an IGBT, plays a central role in regulating voltage. During operation, the switch can either be closed or open depending on the output demand. When closed, current flows through the circuit, inducing a magnetic field within the inductor. This process stores energy in the inductor, preparing it for voltage enhancement. Conversely, upon opening the switch, the polarity of the magnetic field reverses, causing the inductor to release stored energy. This gradual release elevates the voltage level, ensuring that the output voltage across the load or resistor increases irrespective of input voltage.

Diodes are strategically placed within the circuit to prevent backflow, thereby stabilizing the output voltage. This configuration allows the boost converter to efficiently step up the voltage without the need for a transformer, ensuring high efficiency and minimal power loss. Moreover, the input current remains continuous throughout the operation, further enhancing efficiency.

To optimize performance, control mechanisms such as Proportional-Integral (PI) controllers and Maximum Power Point Tracking (MPPT) algorithms are employed. These mechanisms regulate the operation of the IGBT switch, ensuring that the boost converter operates at its maximum efficiency while maintaining voltage stability and meeting the power demands of the grid-tied inverter.

Perturb and observed theorem:



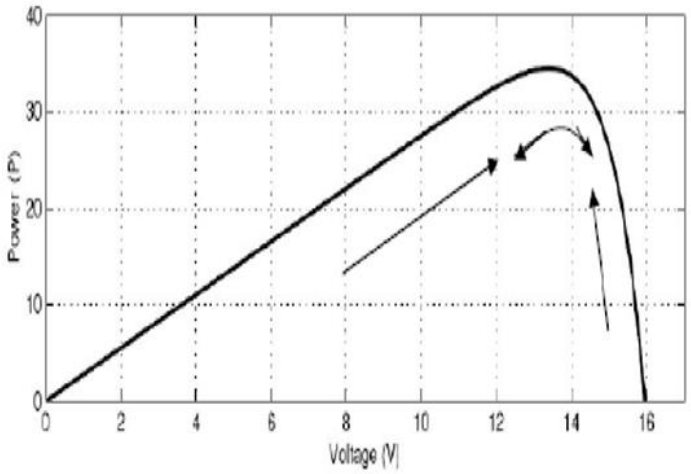
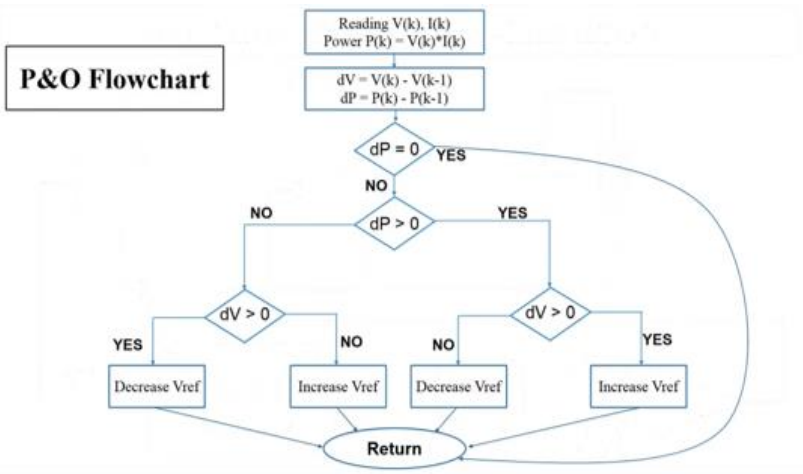


Figure 2:MPPT



The logic of this algorithm and the flowchart are explained in the above figure .The operating voltage of the PV system is perturbed by a small increment of ΔV, and this resulting change in ΔP. If ΔP is positive, the perturbation of the operating voltage needs to be in the same direction of the increment. On the contrary, if ΔP is negative, the obtained system operating point moves away from the MPPT and the operating voltage needs to move in the opposite direction of the increment.

2.2.3 Clarke and Park Tranformation:

The Clarke transform converts the time domain components of a three-phase system (in abc frame) to two components in an orthogonal stationary frame (αβ). The Park transform converts the two components in the αβ frame to an orthogonal rotating reference frame (dq).

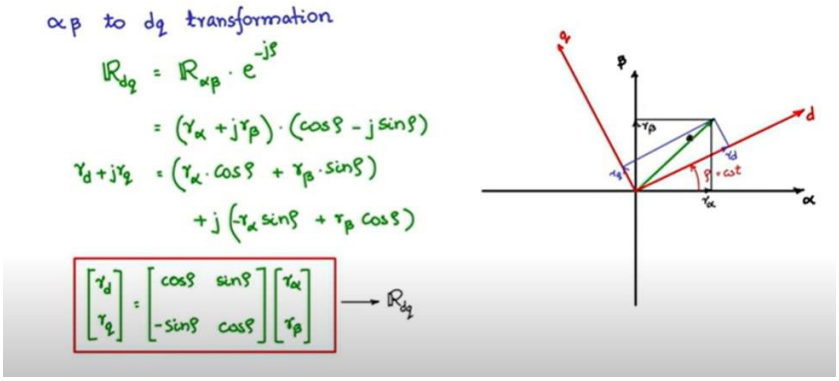


Figure 3:alpha beta to dq transformation

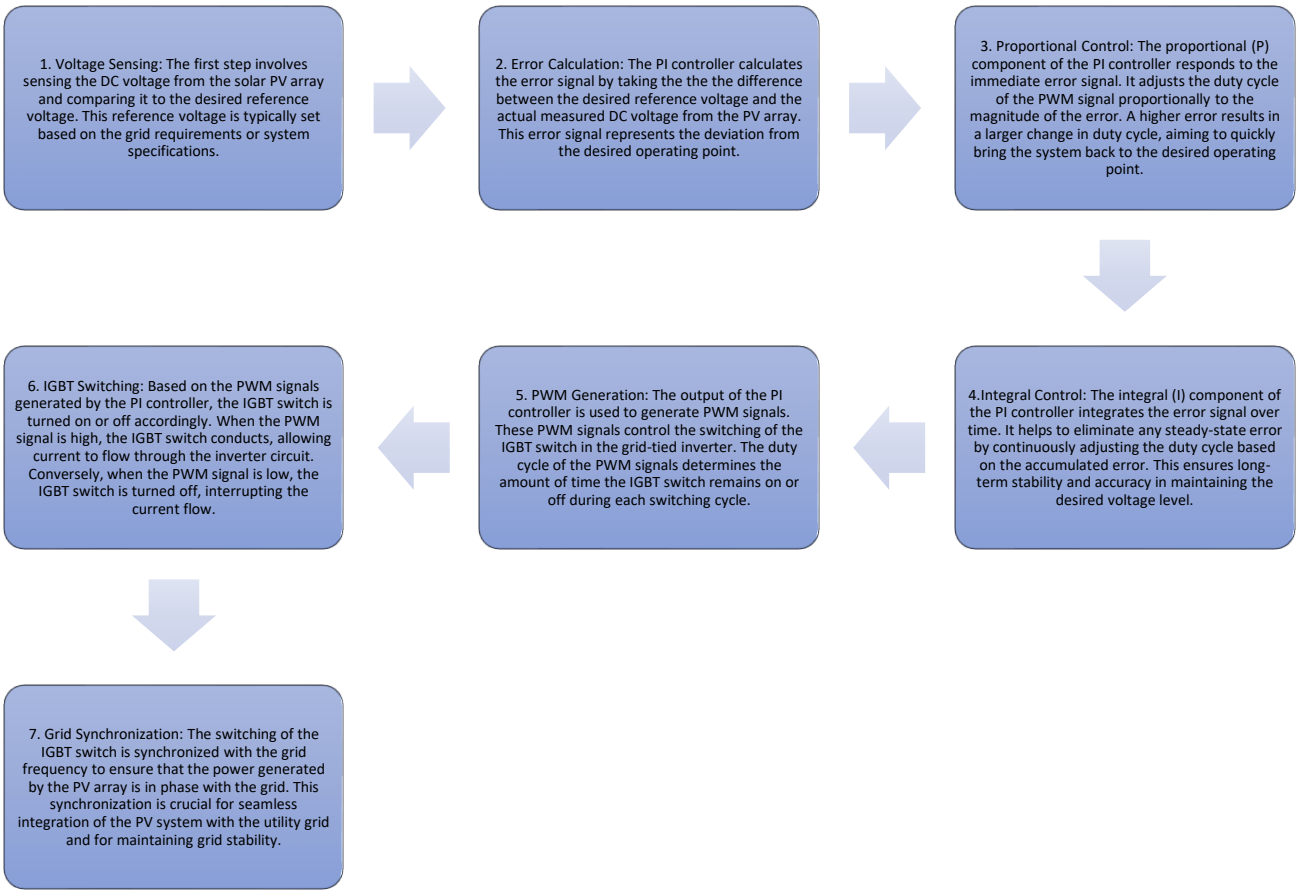
2.2.4 PLL:

The dqPLL is designed according to the Clarke and Park transformation which converts the natural abc reference frame into the synchronous dq-reference frame. To acquire the phase of the input voltages, the q-component of the positive sequence voltage tracks a zero reference through a PI controller, the loop filter . As a result, under ideal voltage conditions, θ(dqPLL) becomes equal to the phase angle of the three-phase voltage and component vd perfectly tracks the magnitude of the positive sequence voltage **v**+. Since the synchronous frame is rotating with the positive angular speed, the dqPLL works accurately for balanced grid faults. It fails to track the phase angle when an unbalanced fault occurs. This is because of the presence of double-line frequency oscillations induced by the negative sequence components **v**- that disturb the dq-components resulting in the mismatch of vH from the positive sequence magnitude |**v**+| .In addition, the $dqPLL$ cannot work for harmonically distorted three-phase voltages.

2.2.5 PV Inverter:

A grid-tied PV inverter is a device that converts the direct current into alternating current.The grid-tied PV inverter design mechanism employs a Proportional-Integral (PI) controller to generate Pulse Width Modulation (PWM) signals for controlling the on-off operation of the IGBT switch. Grid tie inverters typically operate at 10kHz switching frequency and therefore, require ultrafast IGBTs with balanced switching and conduction losses.

The mechanism:



2.2.6 Grid-tied inverter and STATCOM harmonic Filter design:

Pv array gives us dc output which is then boost up by dc dc converter and then it is passed through grid connected pv inverter that basically converts dc to ac but there is a problem regarding harmonic component . Thats why harmonic LCL filter is used to compress higher harmonic component. We know that if frequency increases then reactance of capacitor decreases so higher harmonics is passed through that path and for inductor branch, reactance increases with increase of frequency so only desired low frequency is passed to the grid and thus the higher harmonic problem is solved. Damping resistor is added in series with capacitor and inductor to provide stability in resonant condition but it has a trade off of causing attenuation .

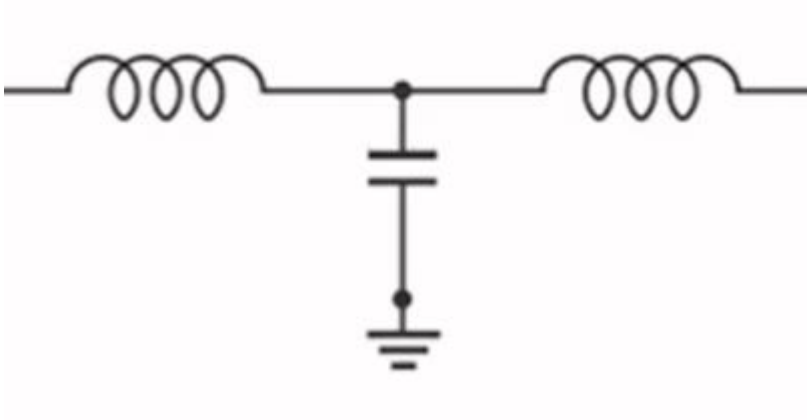


Figure 4:LCL filter

The harmonic filter for the grid-tied inverter is designed based on solar power ratings, switching frequency of the inverter, dc-link voltage, grid voltage, and grid frequency. The following equation used for the design of harmonic LCL filter for grid-tied inverter ,

$$\omega_g = 2 \times \pi \times f_g = 2 \times \pi \times 50 = 314.1593\ rad/s$$

$$\mathrm{Z_b} = \frac{V_g^2}{(\frac{P}{3})} = \frac{(230)^2}{(\frac{100.345 \times 1000}{3})} = 1.5870$$

$$\mathrm{C_b} = \frac{1}{\omega_g \times \mathrm{Z_b}} = \frac{1}{314.1593 \times 1.5870} = 0.002$$

$$I_{max} = \frac{P \times \sqrt{2}}{3 \times V_g \times 0.9} = \frac{100.345 \times 1000 \times \sqrt{2}}{3 \times 230 \times 0.9} = 227.7317A$$

$$\Delta I_{max} = \frac{10}{100} \times I_{max} = \frac{10}{100} \times 227.7317 = 22.7732A$$

$$\mathrm{L_1} = \frac{V_{out}}{6 \times F_{SW} \times \Delta I_{max}} = \frac{700}{6 \times 10000 \times 22.7732} = 0.6mH$$

$$C_g = 0.05 \times C_b = 0.05 \times 0.002 = 100.29\ \mu F$$

$$L_2 = \frac{\sqrt{\frac{1}{0.2^2}} + 1}{C_g \times (2 \times \pi \times F_{SW})^2} = \frac{\sqrt{\frac{1}{0.2^2}} + 1}{100.29 \times 10^{-6} \times (2 \times \pi \times 1000)^2} = 15uH$$

$$\omega_{res} = \sqrt{\frac{L_1 + L_2}{L_1 \times L_2 \times C_g}}$$

$$\omega_{min} = 10 \times f_g$$

$$\omega_{max} = 0.5 \times F_{SW}$$

here, fg is the grid frequency, Vg is the grid voltage per phase,ωg angular frequency in radians of the grid, Zb is per phase impedance of the harmonic filter, Cb is the harmonic filter base capacitance, $Imax$ is the harmonic filter maximum current rating, $\Delta Imax$ is the harmonic filter maximum current ripple of the filter inductor, $L1$ is the input side filter inductor of the harmonic filter, $L2$ is harmonic filter output side filter inductor, Cg is harmonic filter capacitor of the, ωres is the resonance frequency of harmonic filter in radians, ωmin is the minimum frequency limit of the harmonic filter and ωmax is the maximum frequency limit of the harmonic filter.

Hamonic Filter design methodology for STATCOM is same as grid tied inverter .

2.2.7 Capacitor Bank:

Capacitor banks are used to help regulate power factors in electrical systems and can be installed parallel to a load to reduce the amount of reactive power flowing through lines. This improves the power factor by reducing the phase difference between voltage and current. But the drawback is that it is not dynamic so it can't sense the variable demand of the load rather gives a fixed compensation. The rest compensation will be done by statcom which will be observed in the results and outputs.

2.2.8 STATCOM:

A **static synchronous compensator** (**STATCOM**), is a shunt-connected Voltage Source Converter and can act as either a source or sink of reactive AC power which is tied to a transmission line.in this experiment , we used vsc 3 phase 3 wire statcom.

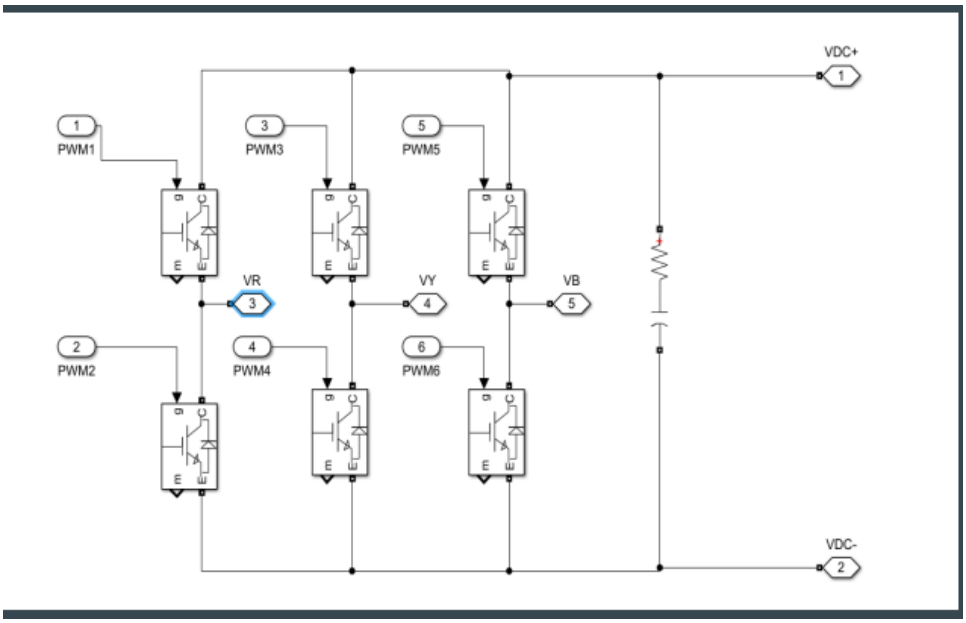


Figure 5:Statcom Switching Circuit

Components:

DC bus link capacitor,IGBT switches

The major attributes of STATCOM:

- Faster response
- Requires less space
- It can be interfaced with real power sources such as battery, fuel cell

STATCOM could have many topologies, but in most practical applications it employs the DC to AC converter, which can also be called a Voltage Source convertor (VSC) in 3-phase configuration as the primary block. The basic theory of VSC is to produce a set of controllable 3-phase output voltages at the fundamental frequency of the AC bus voltage from a DC input voltage source such as a charged capacitor or a DC energy supply device. By varying the magnitude and phase angle of the output voltage and current, the system can exchange reactive power between the DC and AC buses, and regulate the AC bus voltage.

The STATCOM regulates the voltage magnitude at its terminals by controlling the amount of reactive power injected into or absorbed from the power system. When system voltage is low, the STATCOM generates reactive power The charged capacitor provides a dc voltage to the converter, which produces a set of controllable three phase output voltages with frequency of the ac power system. By varying the amplitude of the output voltage , the reactive power exchange between the converter and the AC system can be controlled. If the amplitude of the output voltage is increased above that of the AC system ,a leading current is produced.

2.2.9 Circuit Diagram:

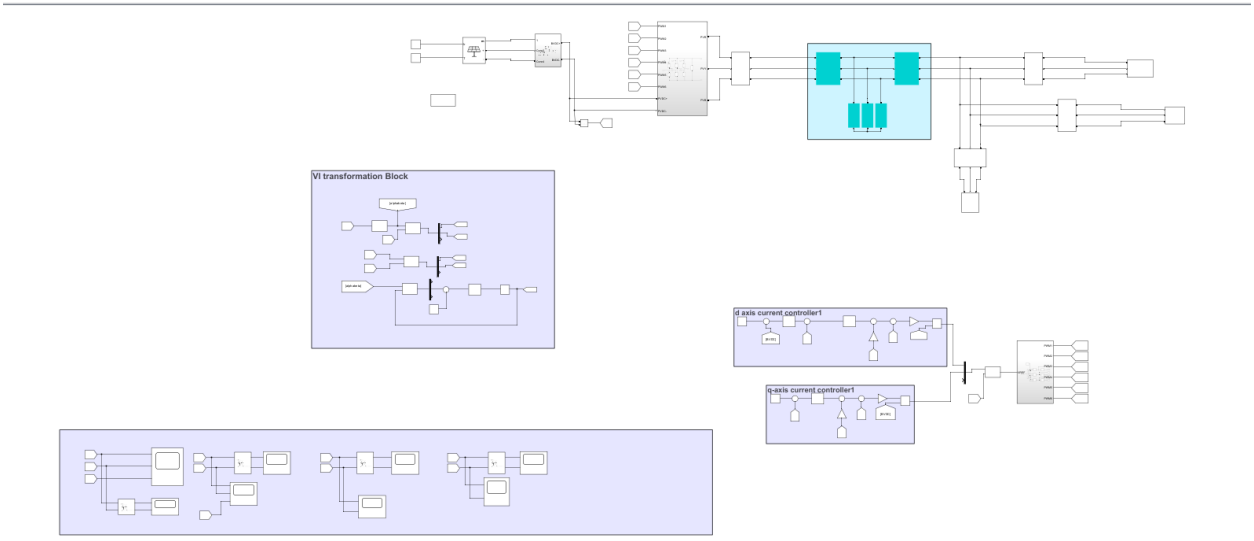


Figure 6:Capacitor Bank based Grid connected PV cell

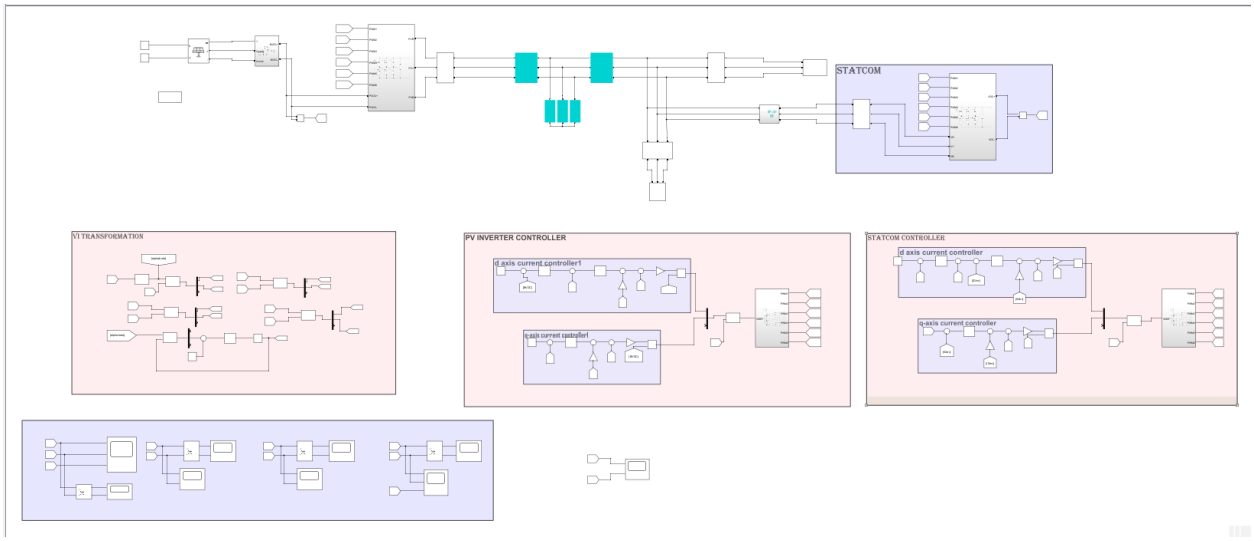


Figure 7:STATCOM based Grid connected PV cell

2.2.10 Simulation model:

Done in matlab Simulink.

3 IMPLEMENTATION:

The whole project is designed based on Simulink model and the steps for modeling in the Simulink model given below:

3.1 Description:

The MATLAB SIMULINK model consists of following sub circuits and model:

PV Array:

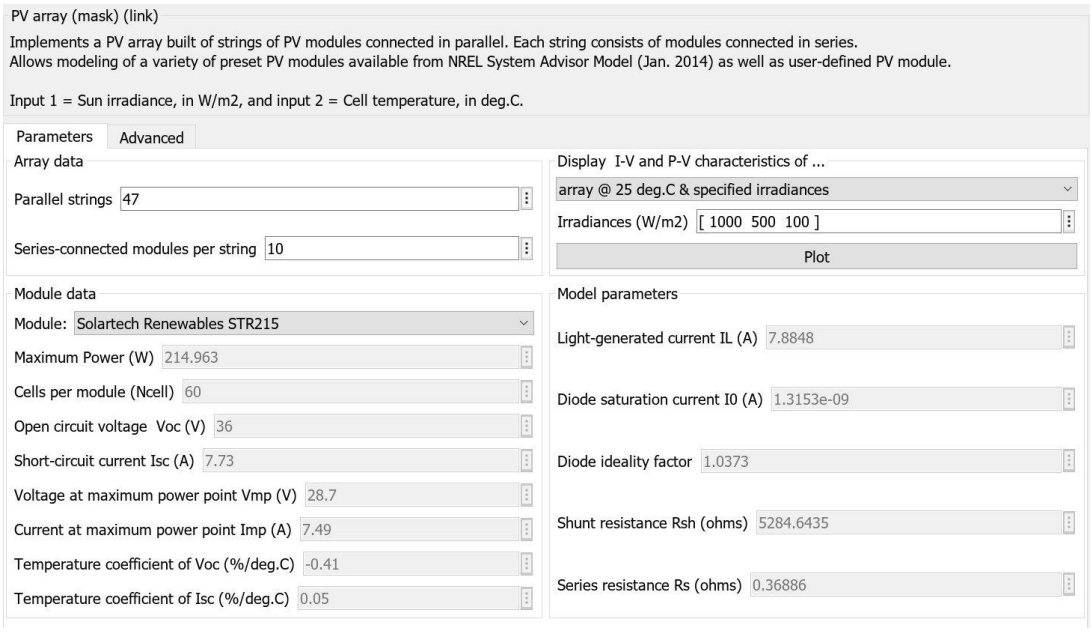


Figure 8:PV ARRAY

We used Solartech Renewable STR215 module in Simulink for modelling the PV array.

The solar PV array exhibits varying maximum power outputs:

- 9.72 kW at 100 W/m² and 25°C
- 50.75 kW at 500 W/m² and 25°C
- 100.345 kW at 1000 W/m² and 25°C.

3.1.1 DC -DC Boost Converter:

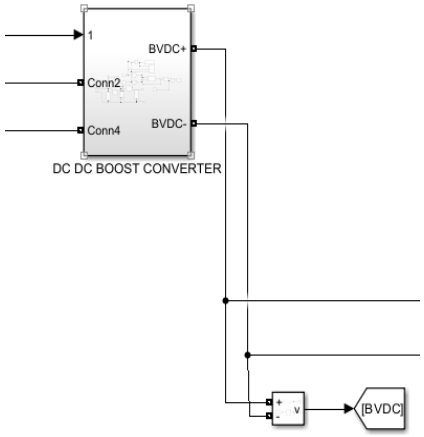


Figure 9:Block for DC DC Boost Converter

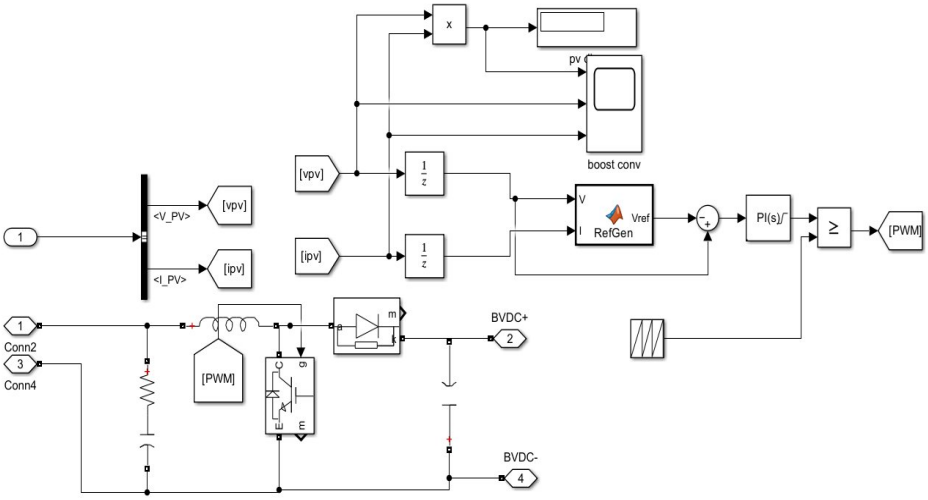


Figure 10:Controller circuit of DC DC Boost Converter

Duty cycle=1-(Vin/Vout) . If Vin is low then duty cycle increases so switch is close for more time which eventually stores more energy that is used to enhance output voltage level. Here output voltage is compared with a carrier signal which produces a pwm with the help of the error calculated by the pi controller block and it controls the duty cycle of the switch and ensures the output voltage of 700v.

MPPT code :

```
function Vref = RefGen(V,I)


Vrefmax=363;
Vrefmin=0.0;
Vrefinit=300;
deltaVref = 1;
persistent Vold Pold Vrefold;

dataType = 'double';

if isempty (Vold)
    Vold=0;
    Pold=0;
    Vrefold=Vrefinit;
end
P=V*I;
dV=V - Vold;
dP= P - Pold;

if dP ~= 0
    if dP <0
        if dV <0
             Vref =Vrefold + deltaVref;
        else
            Vref= Vrefold - deltaVref;
        end
    else
        if dV <0
            Vref= Vrefold - deltaVref;
        else
            Vref =Vrefold + deltaVref;
        end
    end
else Vref= Vrefold;
end

if Vref>=Vrefmax | Vref<=Vrefmin

    Vref=Vrefold;
end

Vrefold=Vref;
Vold=V;
Pold=P;
```

3.1.2 PV Inverter:

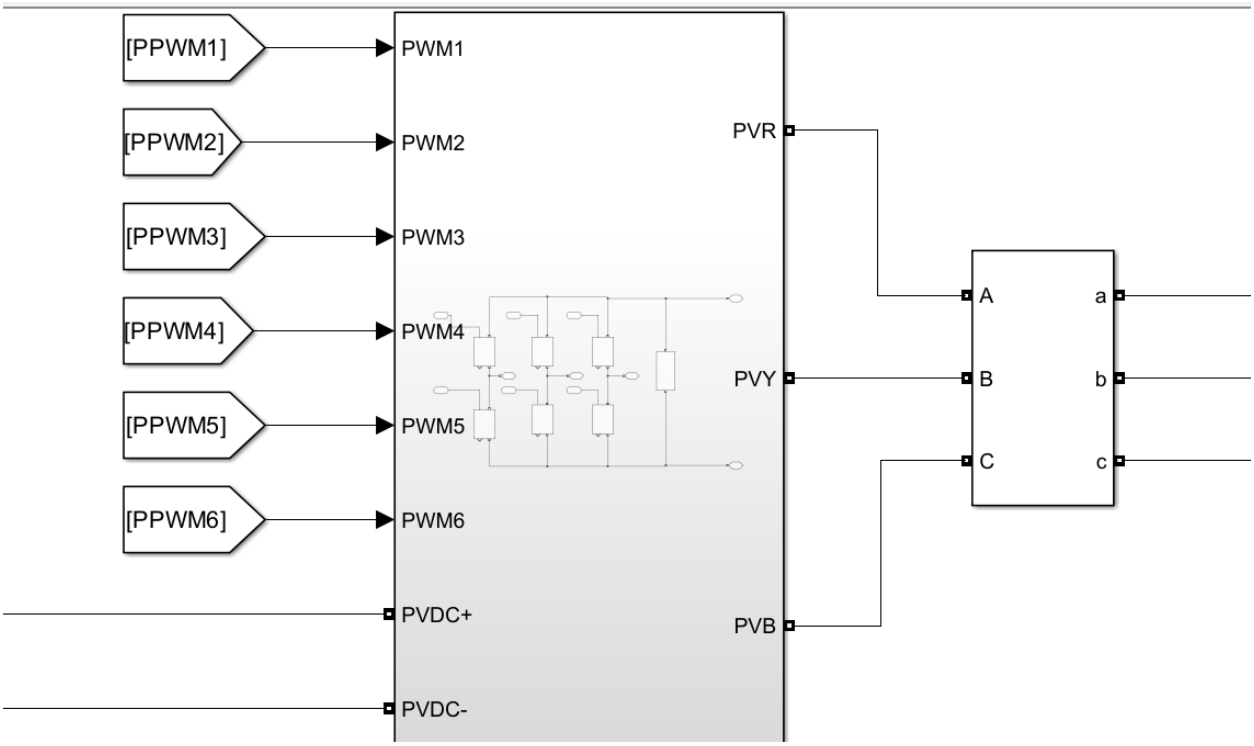


Figure 11:PV inverter block

3.1.3 Voltage Transformation Block & PLL:

Park and Clarke transformation block has been used for converting 3 phase abc grid voltage Vabc ,PV inverter current Iabc , load current ILoad ,Statcom inverter current Iinv to rotating orthogonal d axis , q axis corresponding voltage and current . Also PLL mechanism has been constructed based on PI controller for generating wt .

Pl value has been set such as it compensate the error and generate the desired output.

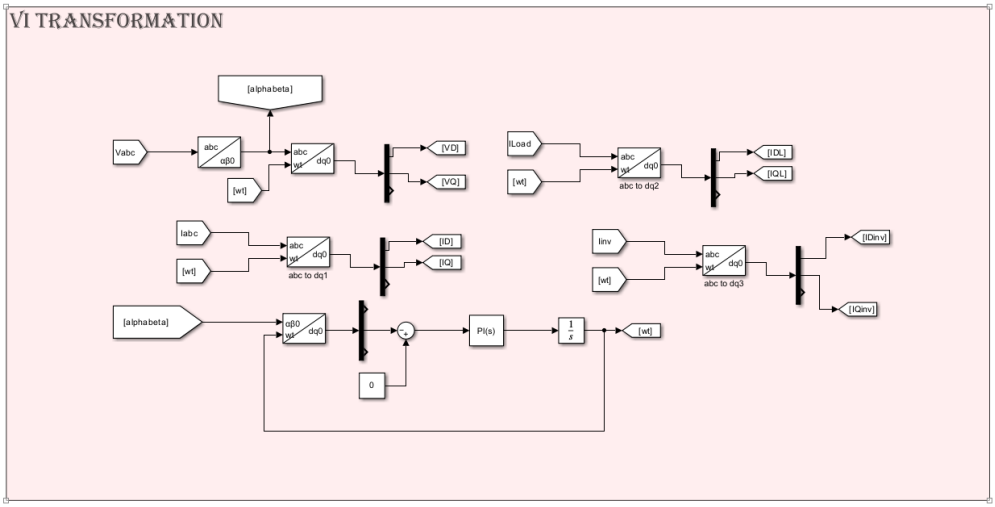


Figure 12:Voltage and current tranformation and PLL block

3.1.4 PI controller For pv inverter :

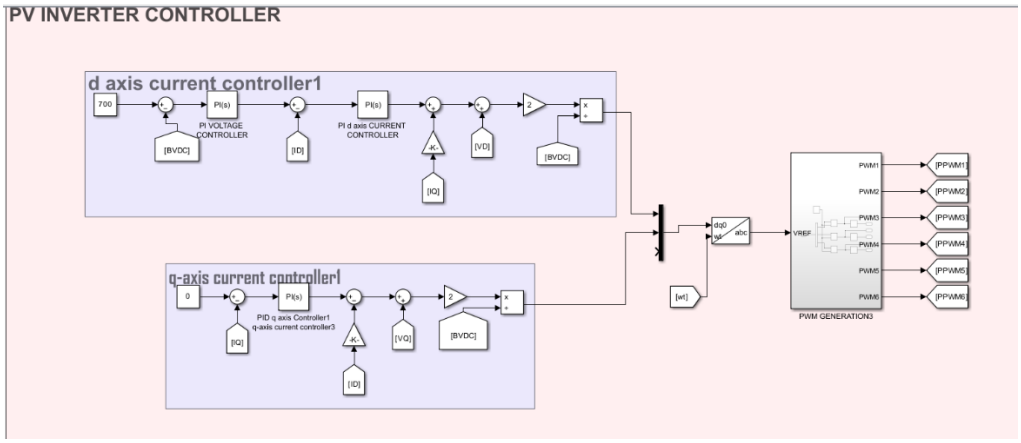
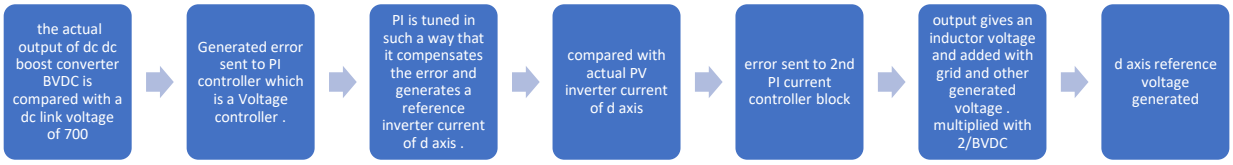
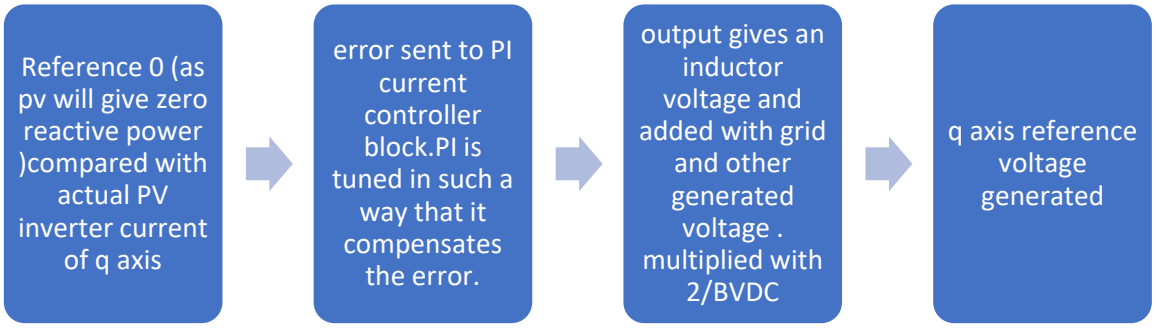


Figure 13:PV INVERTER Controller circuit

D axis current controller:



Q axis controller:



PWM generation block:

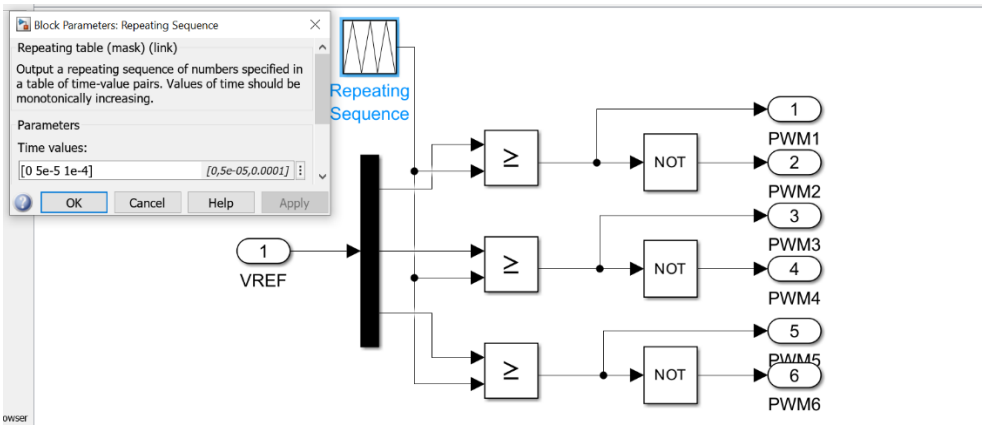


Figure 14:PWM generation bloc

Then performing the inverse park and Clarke transformation the reference 3 phase voltage is compared with a carrier (having switching frequency of 10kHz). And generates six pwm signals.which switches on off the IGBT diodes.

Switching of IGBT and generation of AC signal:

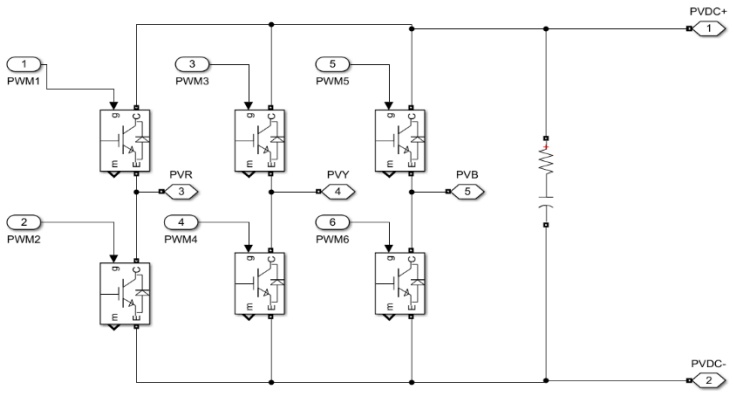


Figure 15:Switching of IGBT and generation of AC signal circuit

- The IGBT switches are being controlled by the PWM signals generated at the PI control.
- The duty cycle is proportional to signal value.
- There are 6 switching sequence.
- There are six modes of operation in a cycle and the duration of each mode is 60°.
- Switches of any leg can't be switch on same time.

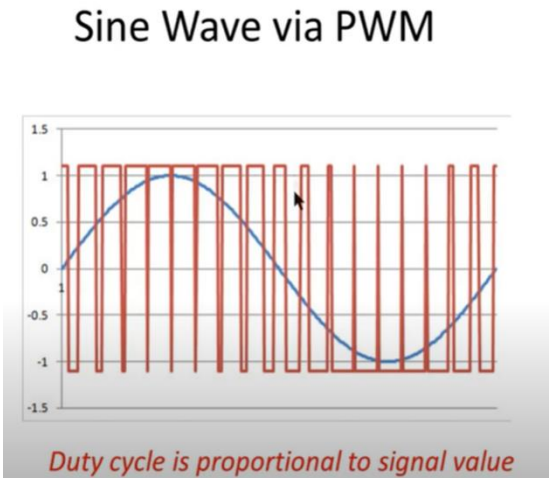


Figure 16: sine wave generation Via PWM

Through the series of on off mechanism a sin wave is generated , through which dc to ac conversion is done.

3.1.5 Grid-tied inverter harmonic Filter design:

The harmonic filter for the grid-tied inverter is designed based on solar power ratings, switching frequency of the inverter, dc-link voltage, grid voltage, and grid frequency. The following equation used for the design of harmonic LCL filter for grid-tied inverter.

The designed value of the input side inductor of the harmonic filter is 0.6 mH, the designed value of the output side inductor of the harmonic filter is 15.15 µH, and the designed value of the capacitor of the harmonic filter is 100.29 µF.

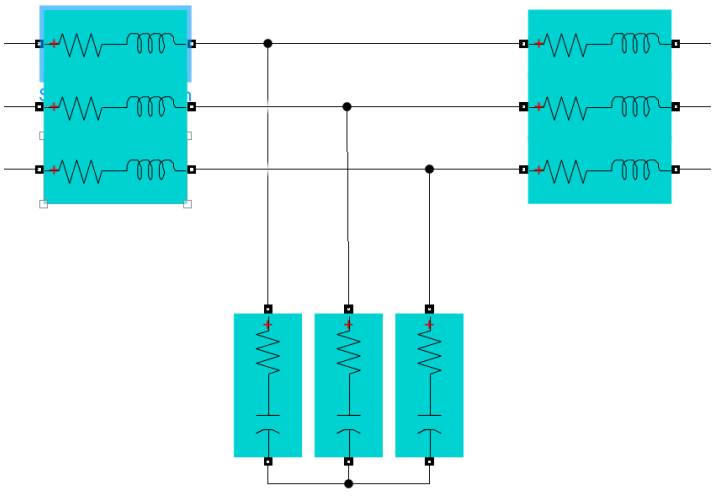
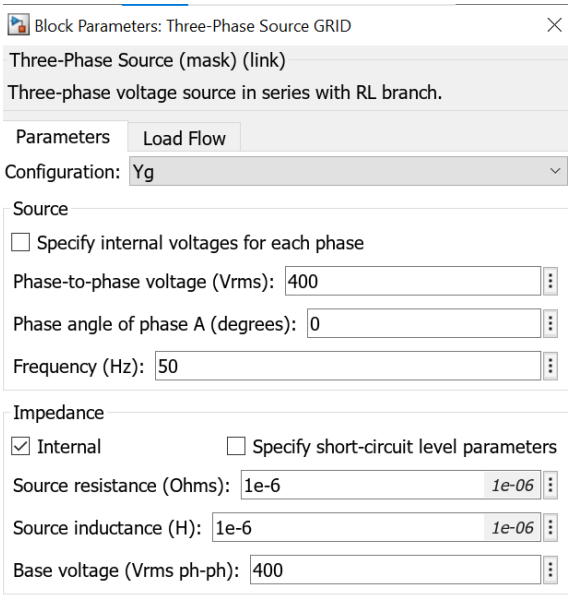


Figure 17:LCL Filter

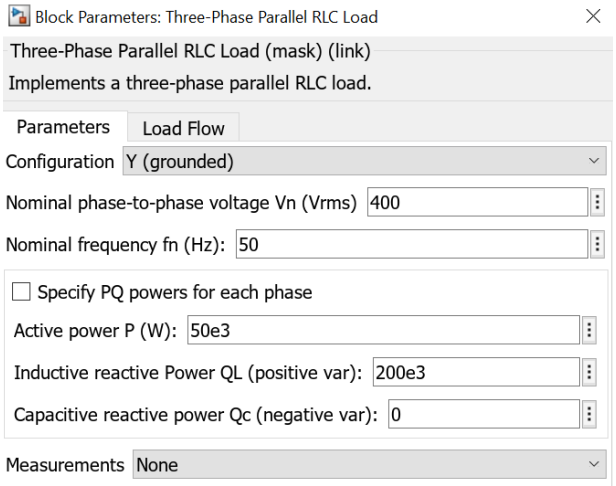
3.1.6

Grid :



Grid's voltage and frequency and other parameters are set within the Grid block

3.1.7 Load:



Load's voltage and frequency and other parameters are set within the Load's block. And we will change the active and reactive power according to our study and preferences

3.1.8 Capacitor Bank:

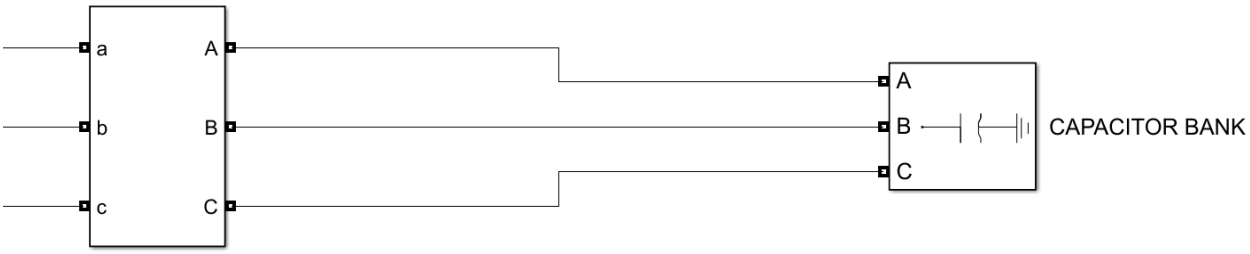


Figure 18:Capacitor bank

Capacitor bank is designed according to our demand of reactive power or how much reactive power we want to supply .

3.1.9 STATCOM:

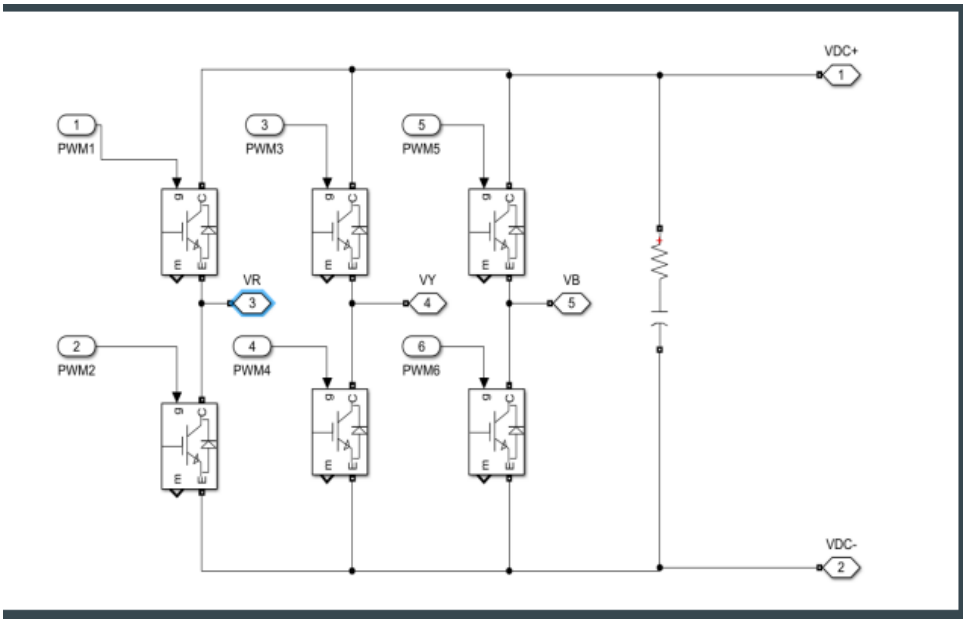


Figure 19:statcom switching circuit

Here, Vdc>= 2*(√2/3) Vl/m ; where m is the modulation index

Assuming 10 percent loss in igbt switches, VDC ref is taken as 800v

This dc link voltage is then converted to 3 phase ac through pwm inverter .

Switching frequency of carrier signal in pwm inverter is taken 10kzH.There are 2 current controllers which give us reference voltage for d and q axis and then it is converted to 3phase abc

frame. After that it is passed through a comparator circuit which produces our desired pwm signal that eventually control the IGBT switches to convert dc to ac.

PI CONTROLLER FOR STATCOM:

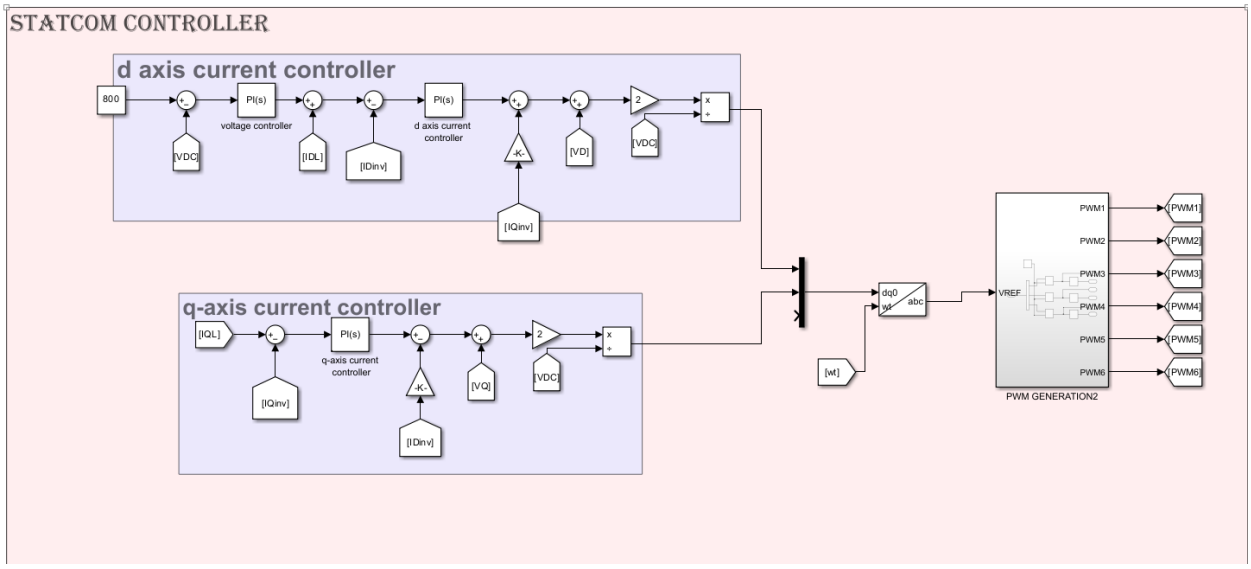
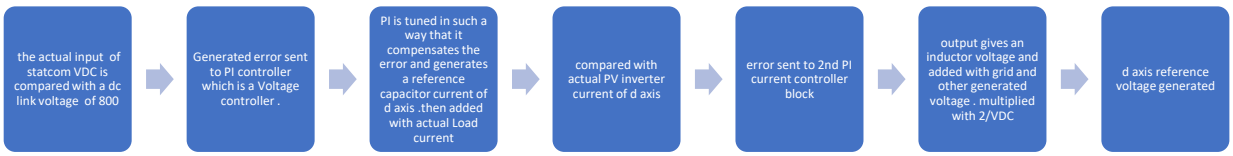
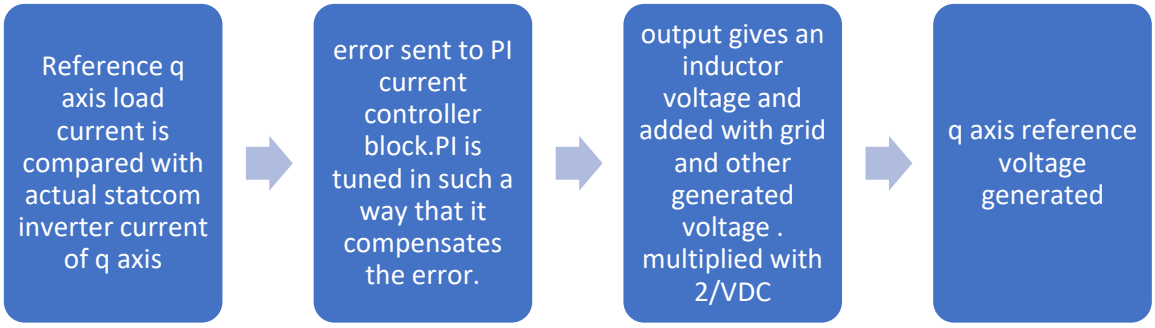


Figure 20:PI CONTROLLER FOR STATCOM:

d axis current controller:



q axis controller:



PWM generation block:

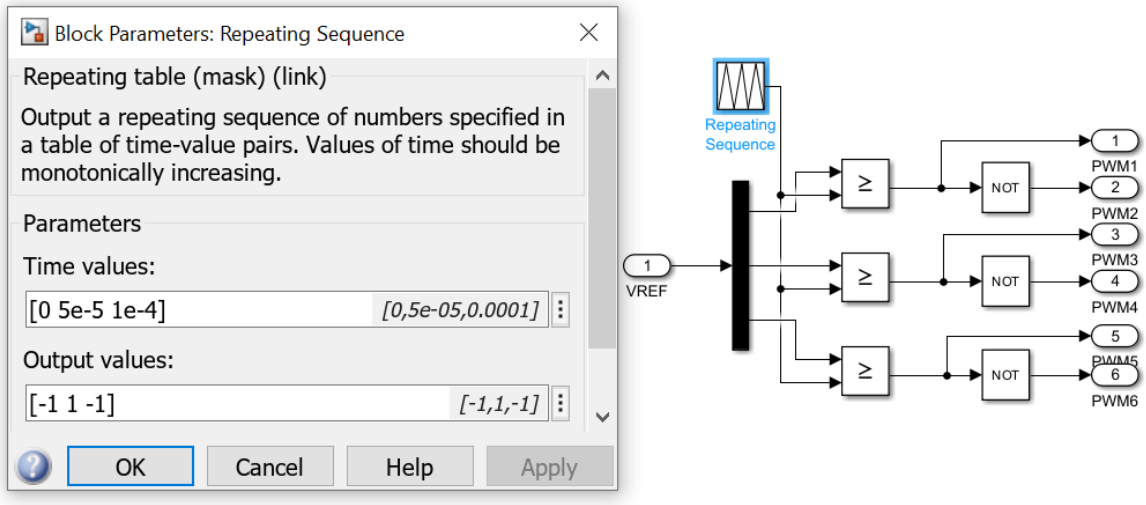


Figure 21:PWM generation block:

Then performing the inverse park and Clarke transformation the reference 3 phase voltage is compared with a carrier (having switching frequency of 10kHz). And generates six pwm signals.which switches on off the IGBT diodes.

Switching of IGBT and generation of AC signal:

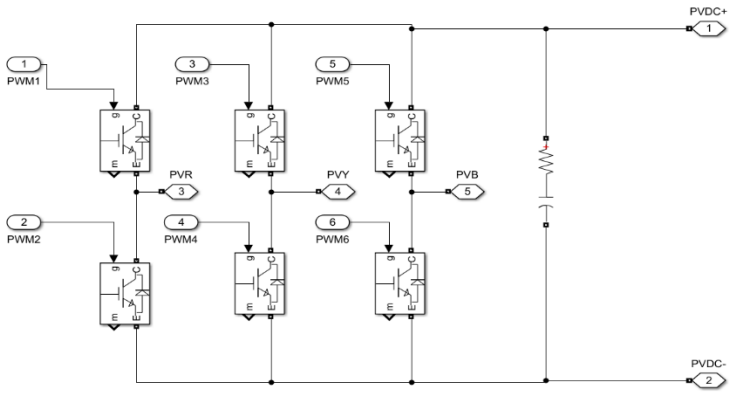


Figure 22:STATCOM IGBT switching circuit

- The IGBT switches are being controlled by the PWM signals generated at the PI control.
- The duty cycle is proportional to signal value.
- There are 6 switching sequence.
- There are six modes of operation in a cycle and the duration of each mode is 60°.
- Switches of any leg can't be switch on same time.

Through the series of on off mechanism a signal wave is generated , through which statcom's dc to ac conversion is done.

3.1.10 Design of harmonic Filter for STATCOM

The harmonic filter for STATCOM is designed based on the reactive power requirement, dc-link voltage, grid voltage, the switching frequency of the inverter, and grid frequency. A similar equation is followed for the design of the harmonic filter of the STATCOM. The designed value of the input side inductor of the harmonic filter in the STATCOM is 0.6 mH, the designed value of the output side inductor of the harmonic filter in the STATCOM is 15.15 µH, and the designed value of the capacitor of the harmonic filter in the STATCOM is 100.29 µF.

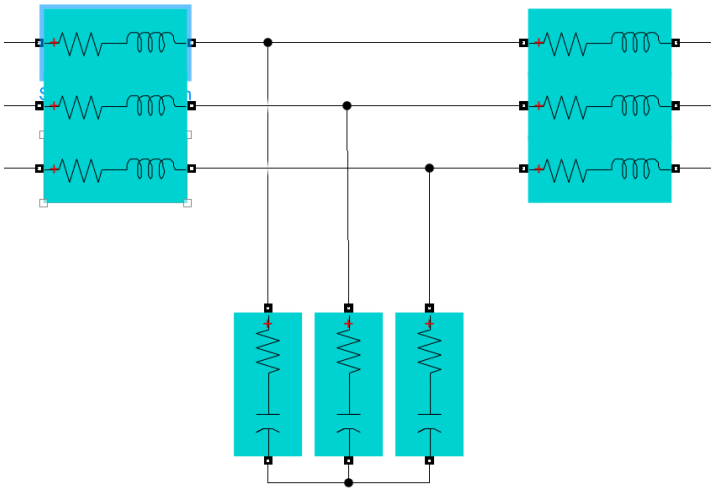


Figure 23:LCL Filter

3.2 Experiment and Data Collection:

Only PV inverter: (absence of capacitor bank and STATCOM)

Case:1

Irradiance :1st set to 1000 then reduced to 500

Load active power :100KW

Reactive power: 400 kvar(inductive)

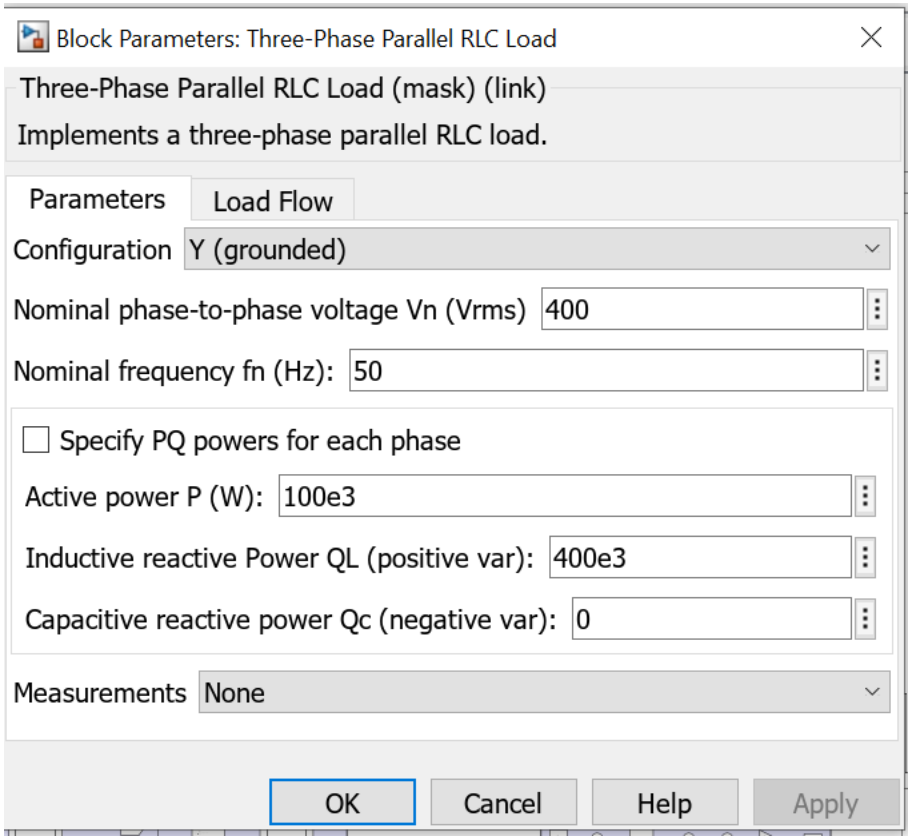


Figure 24:RLC Load

BOOST CONVERTER:

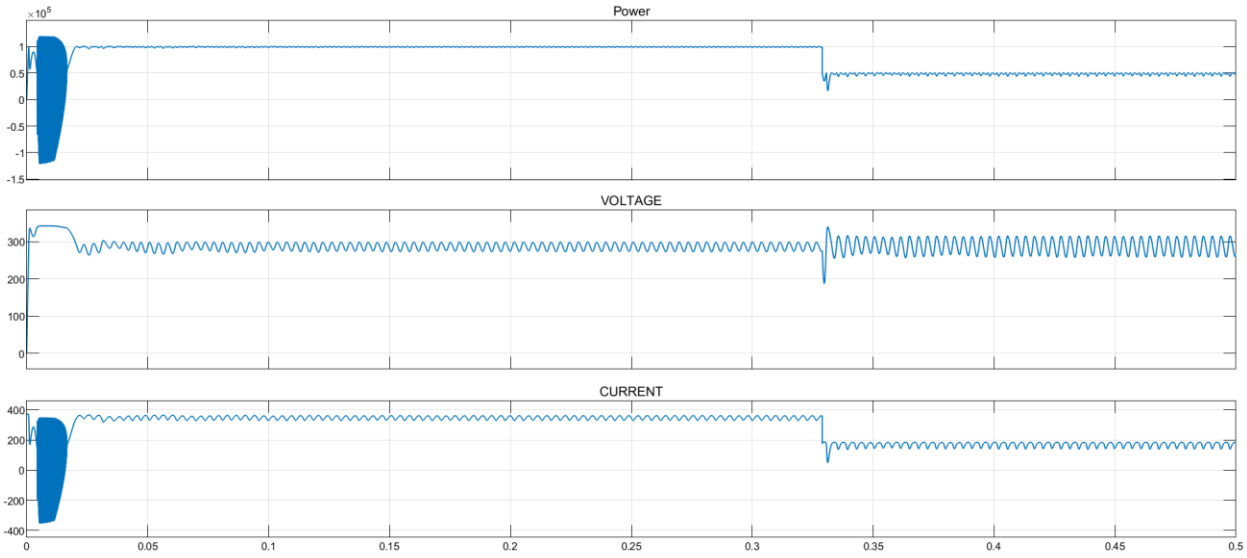


Figure 25: Real power , Voltage , Current of boost converter of PV array

Boost converter's active power 1st gives 100kW and then it reduces to 50KW as irradiance has been changed from 1000 to 500

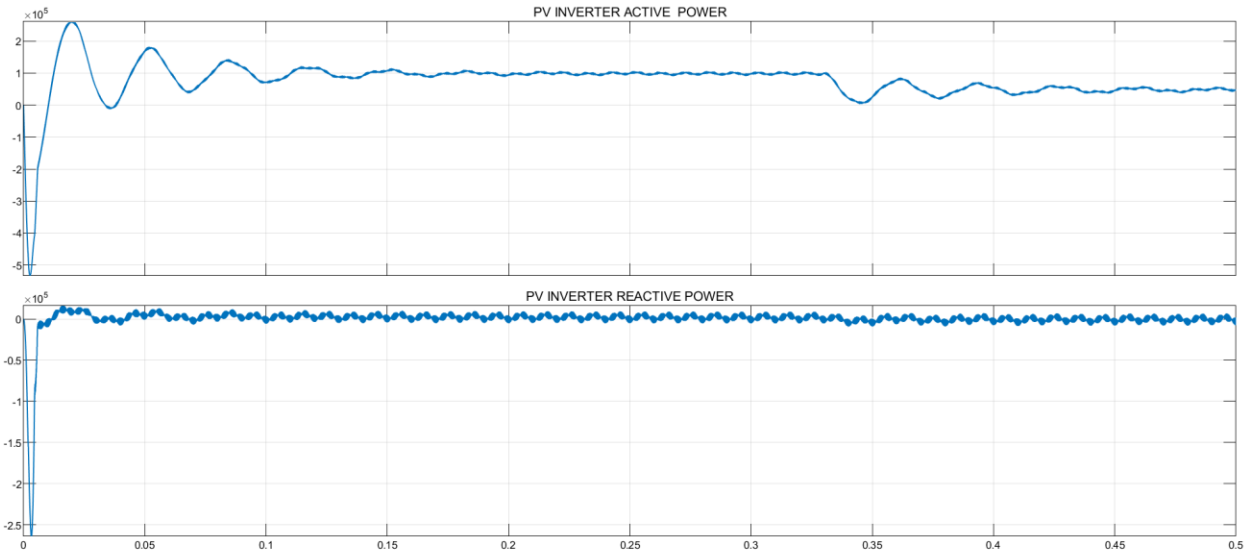


Figure 26: Solar PV inverter real reactive power for case 1

PV inverter's active power 1st gives 100kW and then it reduces to 50KW as irradiance has been changed from 1000 to 500. Again the Reactive power will be zero according to the theory.

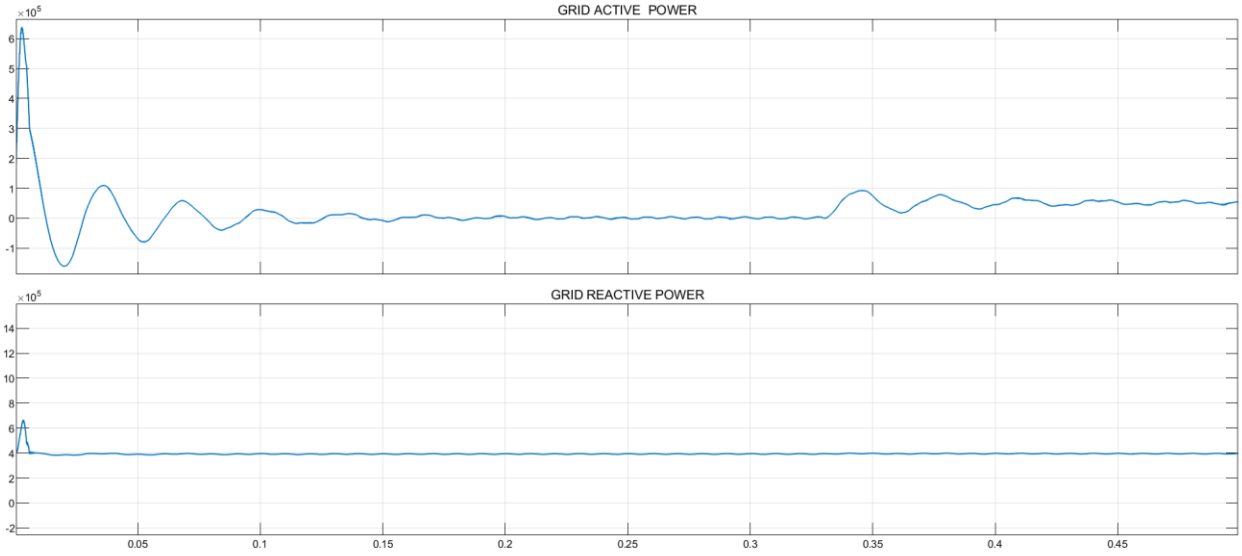


Figure 27:Grid real and reactive power for case 1.

At 1st PV gives full active power and thus grid supplies zero active power. But after irradiance set to 500 , pv will give 50kw and rest 50kW will be given by the grid .

And as there is no reactive power compensation source Grid will supply the whole reactive power according to the load's demand.

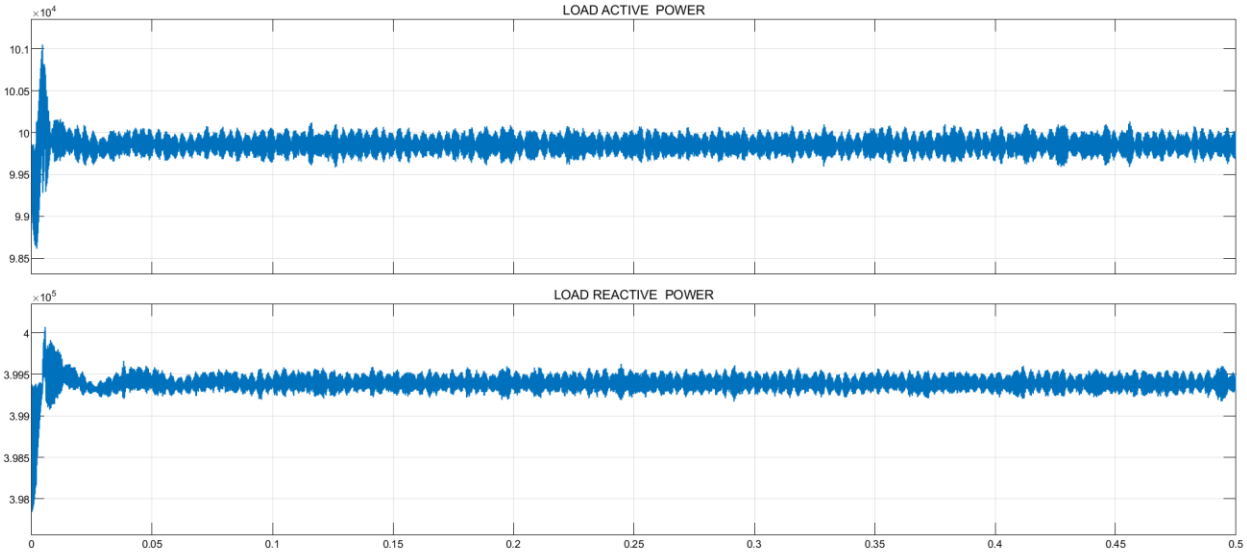


Figure 28:Load real and reactive power for case 1.

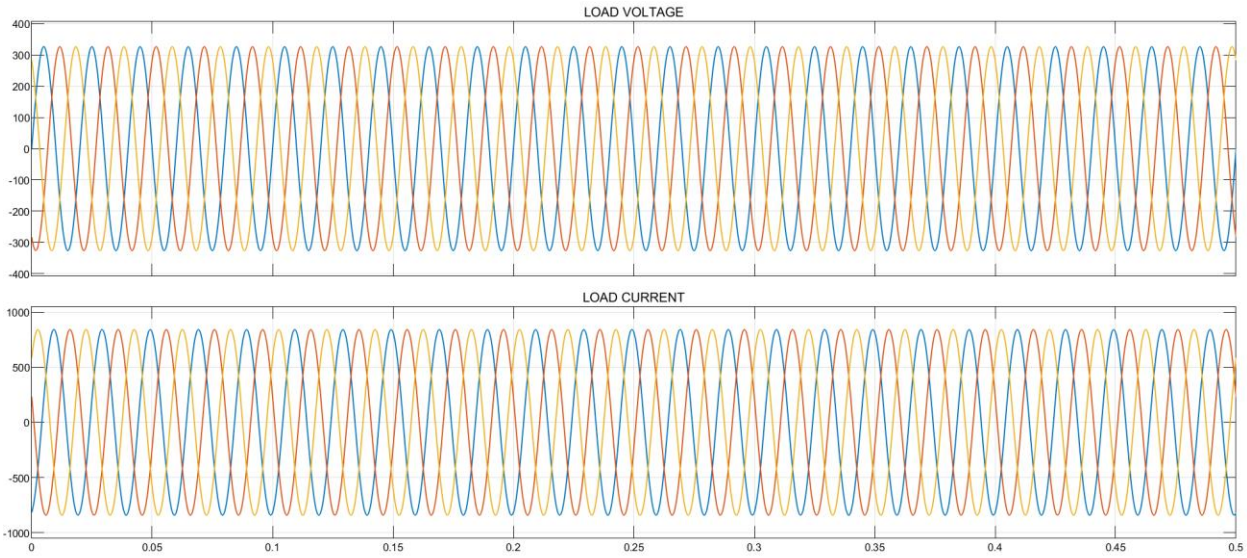


Figure 29: Load voltage and current for case 1

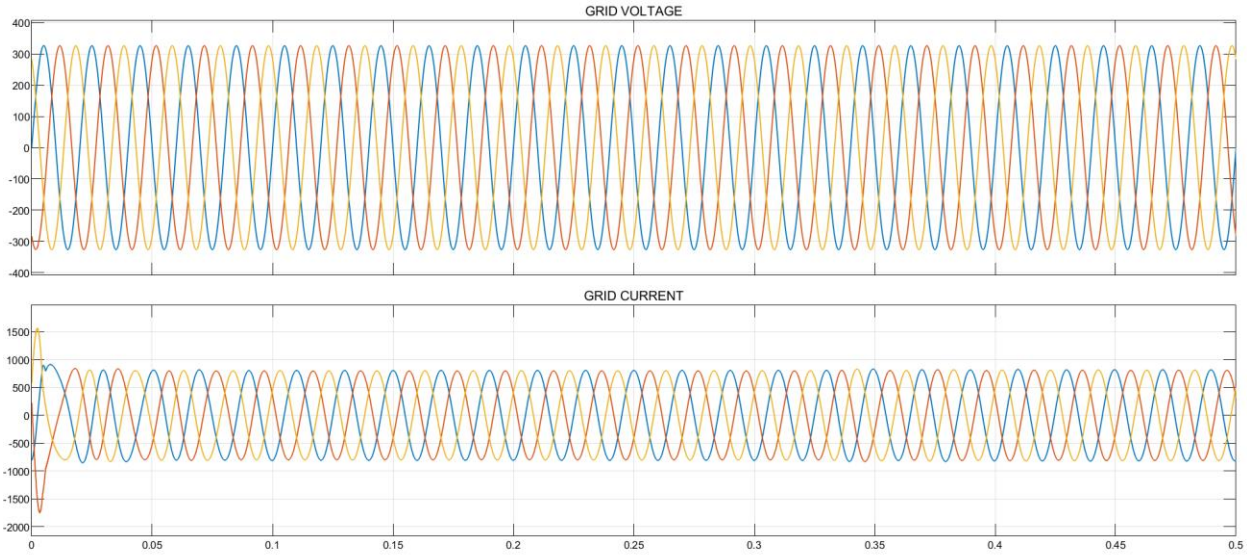


Figure 30: Grid voltage and current for case 1

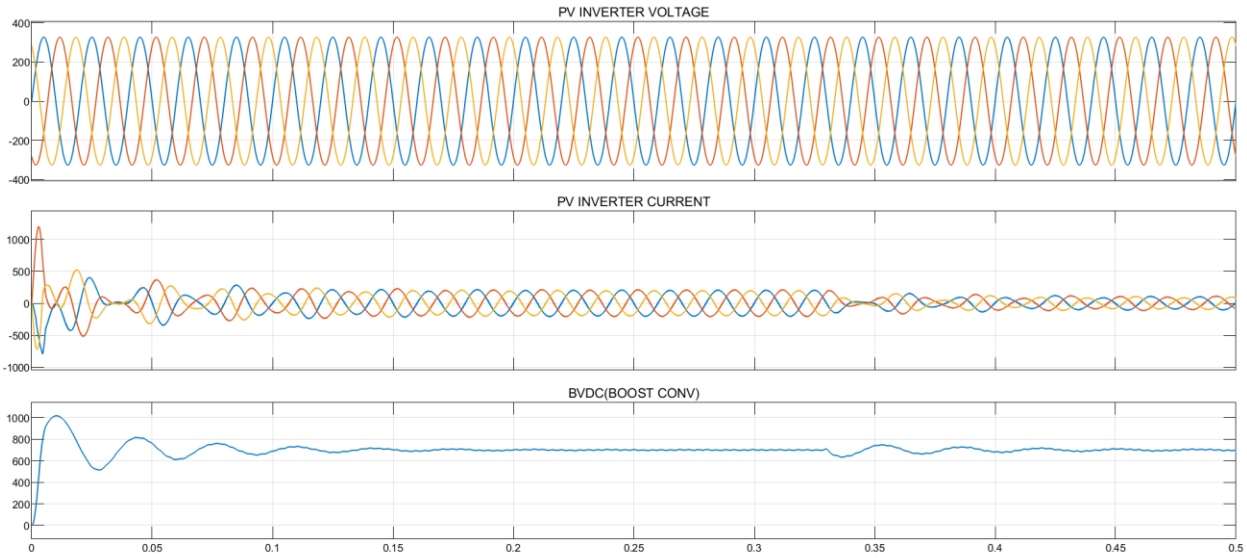


Figure 31: PV inverter dc-link voltage, voltage, and current

Using Capacitor Bank:

Now for compensating reactive power we connect a capacitor bank .

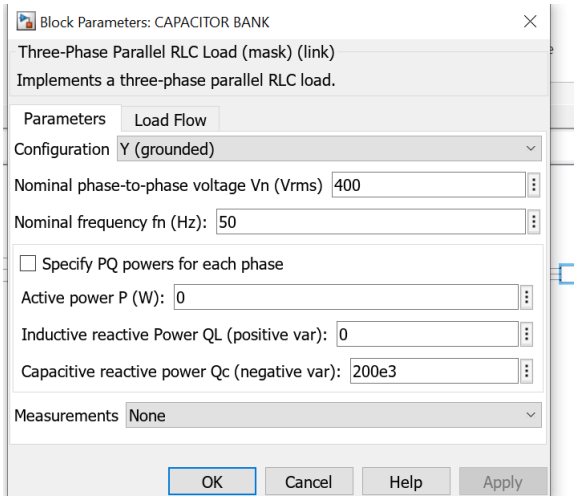


Figure 32:Capacitor bank parameter

Case:2 LOAD's

Active power:100KW

Reactive power:400 kvar

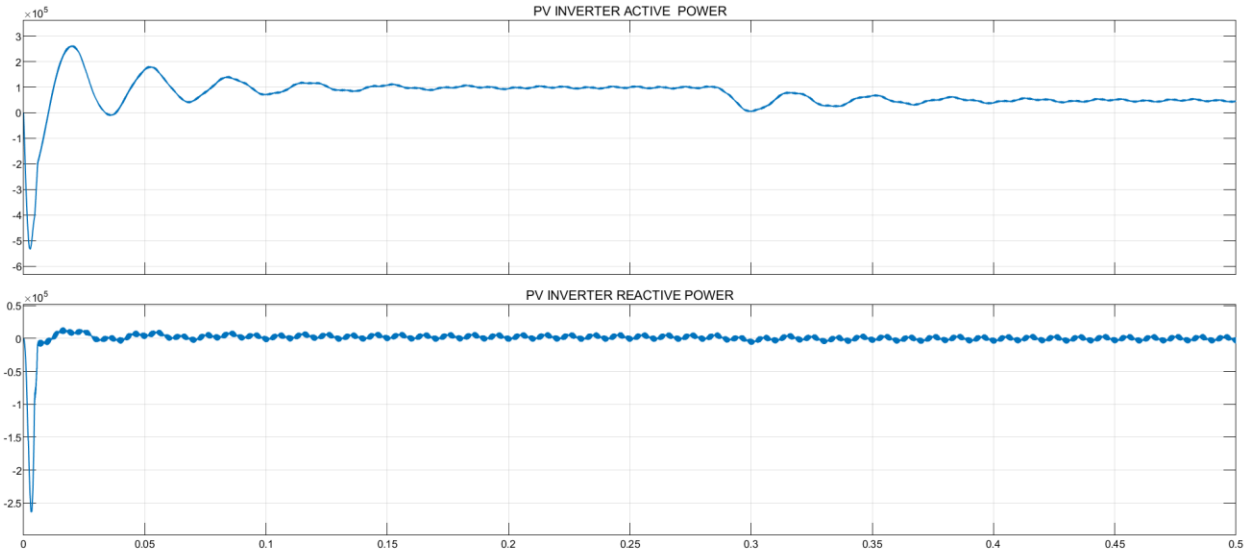


Figure 33:Solar PV inverter real reactive power for case 2

PV inverter's active power 1st gives 100kW and then it reduces to 50KW as irradiance has been changed from 1000 to 500. Again the Reactive power will be zero according to the theory.

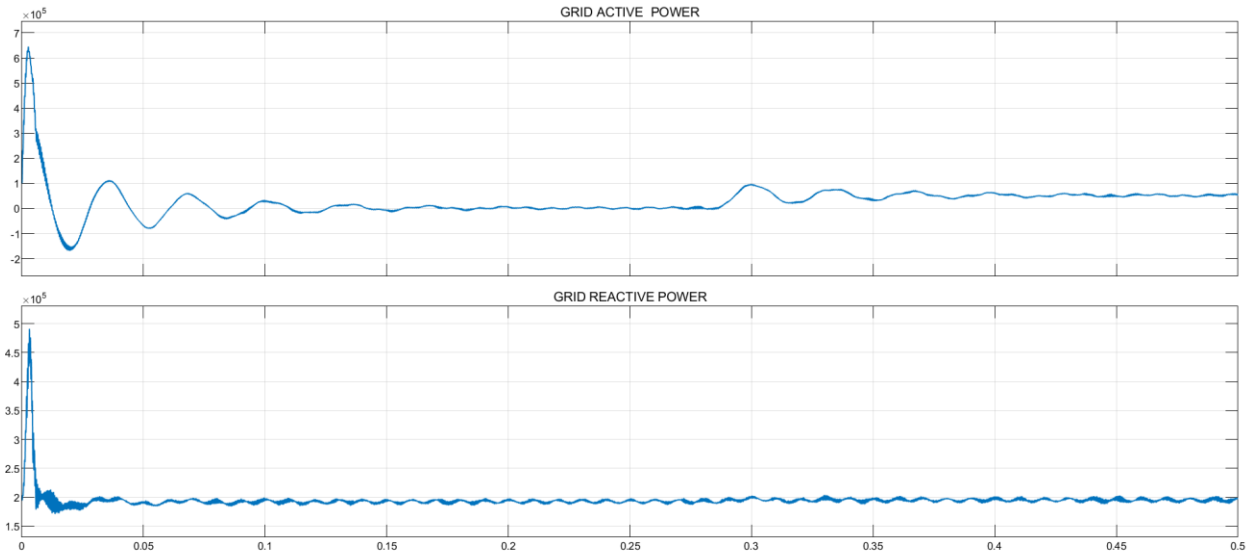


Figure 34:Grid real reactive power for case 2

As we connect a capacitor bank of 200KVAR for reactive power compensation , so bank will suppky 200KVAR and rest 200kVar will be supplied by grid.

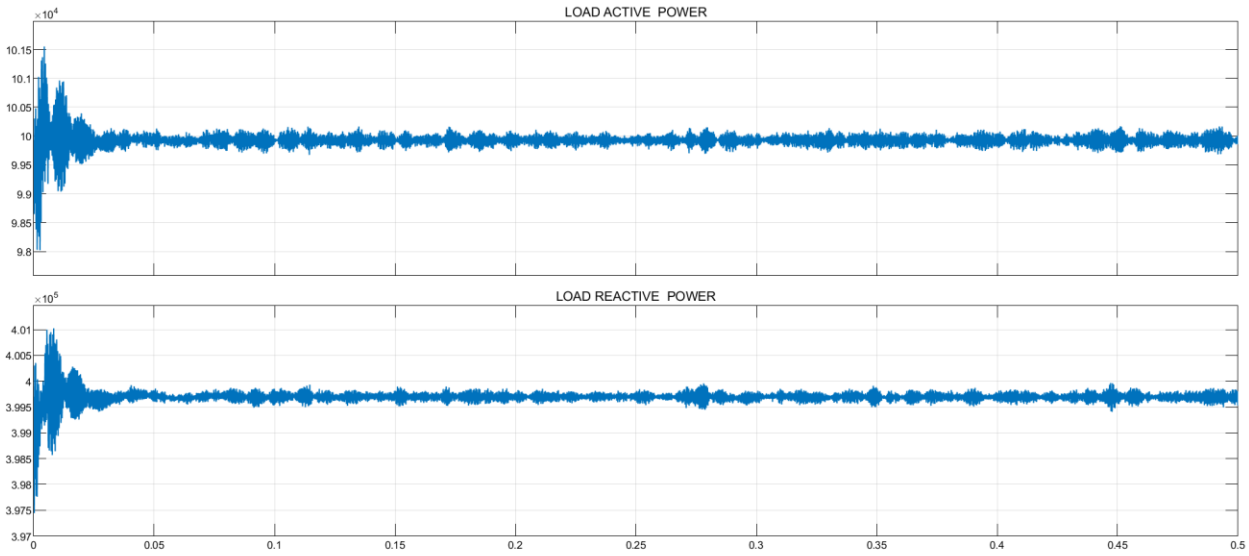


Figure 35:Load real rective power for case 2

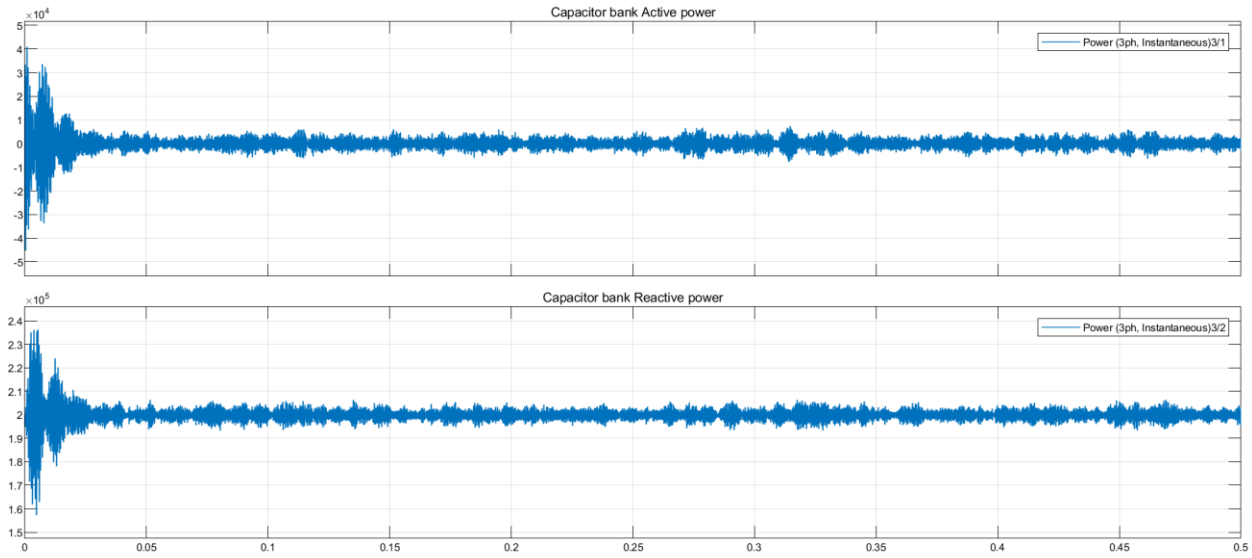


Figure 36: Capacitor Bank real reactive power for case 2

Capacitor bank will give zero active power and will provide the fixed reactive power that has been set to the bank which is 200KVAR.

USING STATCOM

Now for dynamic reactive power compensation we connect a Statcom .

Case 3:LOAD’s

Active power:100KW

Reactive power:200 kvar

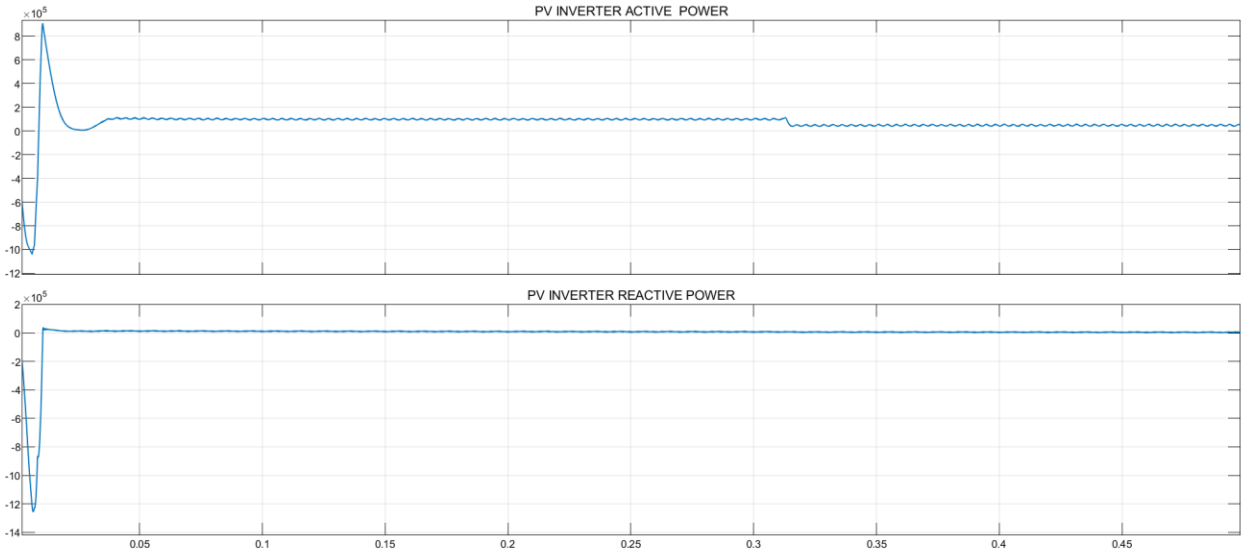


Figure 37:Solar PV inverter real reactive power for case 3

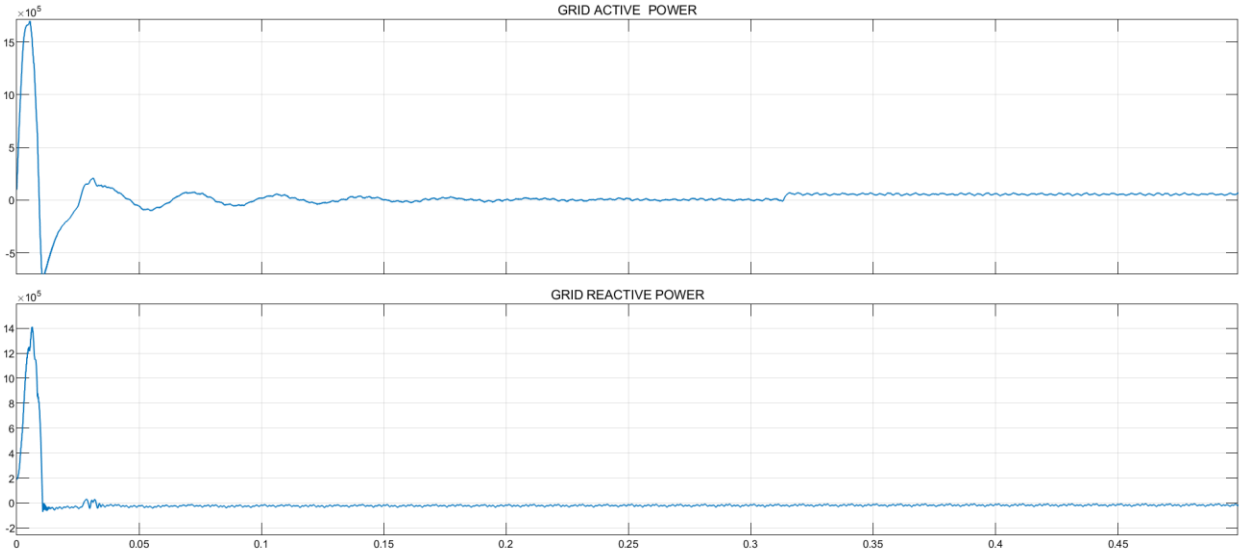


Figure 38:Grid real reactive power for case 3

Here we can see that the reactive power by the grid is zero as whole demand is being full filled by STATCOM.

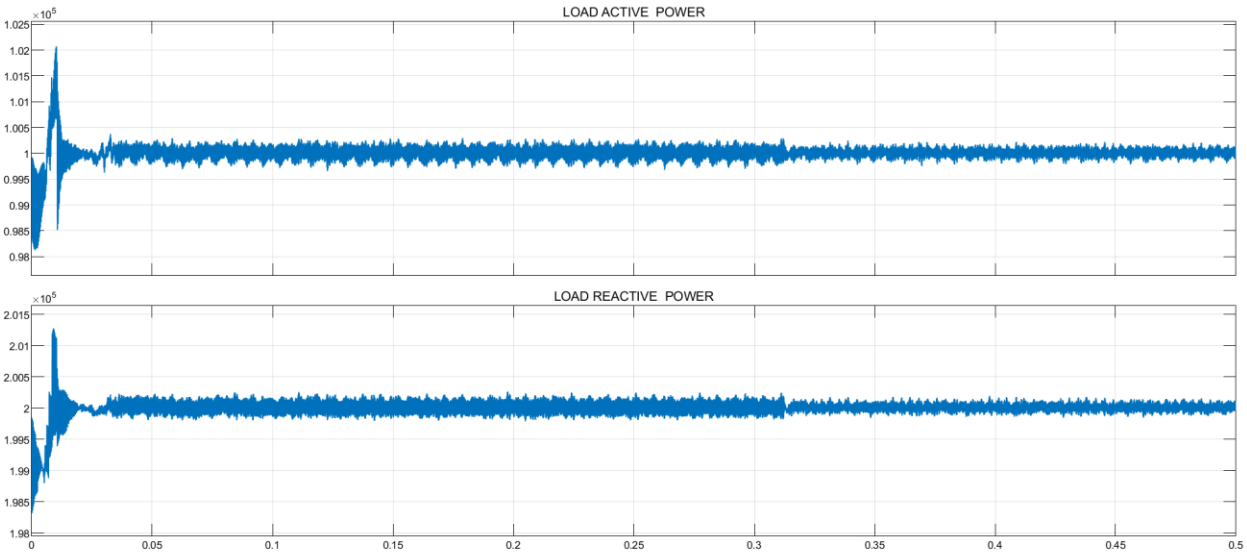


Figure 39:Load real reactive power for case 3

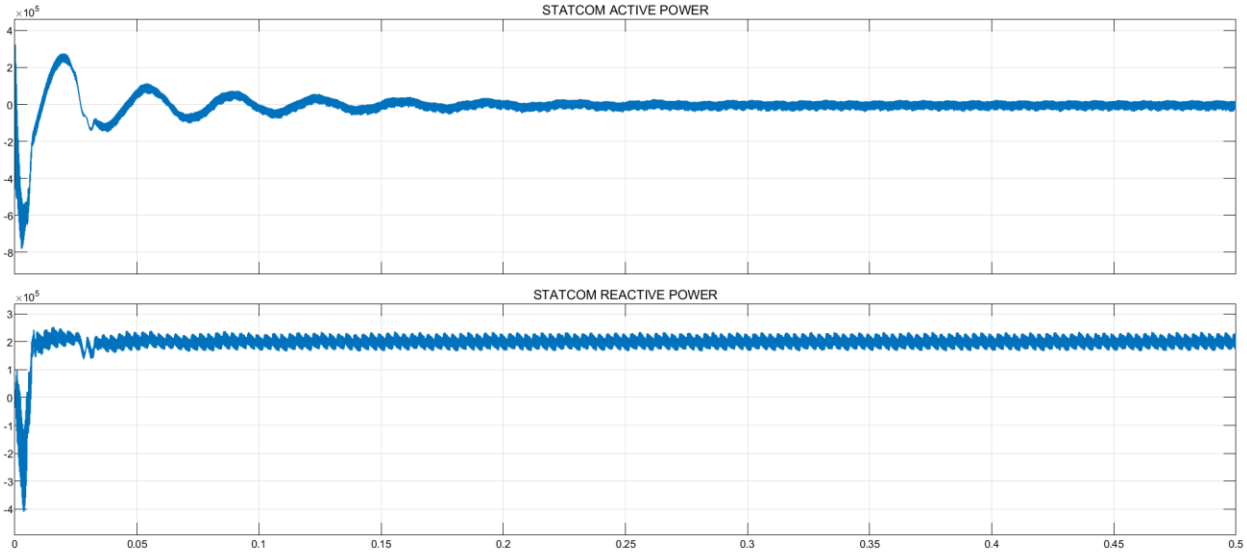


Figure 40:STATCOM real reactive power for case 3

STATCOM will give zero active power and will provide the demanded reactive power by the load that is 200 kvar

3.3 Data Analysis:

No capacitor bank , STATCOM	Reactive power supplied by GRID
Capacitor bank	Provides only the fixed reactive power, rest supplied by GRID
STATCOM	Dynamically adjust the reactive power , no grid supply is need

3.4 Result :

Capacitor banks only provide fixed reactive power, with the remainder supplied by the grid.STATCOM dynamically adjusts reactive power without requiring grid supply.While capacitor banks offer a static solution, STATCOM offers dynamic adjustment capabilities, enhancing grid stability and efficiency.So it is better to use STATCOM for reactive power Compensation.

4 DESIGN ANALYSIS AND EVALUATION:

4.1 Novelty:

The project explores the innovative integration of STATCOM and a Fixed Capacitor Bank in a grid-connected solar PV array system, enhancing reactive power compensation efficiency. As statcom can adjust reactive power dynamically with load's demand.PI controlling mechanism enhance the compensation technique for any kind of load.

4.2 Design Consideration:

4.2.1 Considerations to Public Health and Safety:

Parameters like voltage and current ratings, switching frequency, and control mechanisms are carefully evaluated to ensure optimal system performance and stability. Safety protocols, regulatory compliance, and risk mitigation strategies are prioritized to safeguard public health and ensure operational safety.

4.2.2 Considerations to Environment:

Efforts are made to minimize environmental impact by evaluating energy consumption, emissions, and waste generation, promoting sustainability throughout the project lifecycle.As PV system is basically a natural source of power , environment pollution is minimized through this project.

4.2.3 Considerations to Cultural and Societal Needs:

the project takes into account the values, beliefs, and practices of the local community where the solar PV array system is implemented. This involves engaging with community stakeholders to understand their perspectives, concerns, and aspirations regarding renewable energy projects. Sensitivities related to cultural heritage, land use, and visual aesthetics are carefully considered to ensure that the project design and implementation respect and align with community values.

4.3 Limitations of Tools:

While MATLAB/Simulink is powerful for modeling and simulation in various fields, it can be complex to use, requiring proficiency in MATLAB programming language. It may also have limitations in scalability for large-scale simulations.Besides in Hardware construction, cost can be a significant limitation, as some tools may be prohibitively expensive for individuals or organizations with limited budgets.

5 Individual Contribution of Each Member

5.1 Individual Contribution of Each Member

ID	Model Construction
2006157	PV ARRAY , Filter, boost converter
2006158	STATCOM controller,PLL,voltage current transformation
2006159	Capacitor bank mechanism, STATCOM
2006160	PV ARRAY , Filter,boost converter
2006161	PV inverter , switching of IGBT mechanism
2006163	PV inverter controller, PLL, voltage transformation ,

6 COMMUNICATION:

6.1 Executive summary:

This study explored reactive power compensation in a grid-connected system using advanced STATCOM technology compared to traditional fixed capacitors. The project successfully modeled load dynamics and achieved maximum power point tracking of PV modules through innovative control techniques. However, the high cost of STATCOM systems remains a limitation, suggesting the need for cost-effective alternatives in future implementations. Further research could extend this work to fully grid-connected microgrids, assessing power flows and addressing disruptions. Overall, integrating STATCOMs effectively managed reactive power supply while minimizing voltage disturbances, showcasing their potential for enhancing grid stability and reliability in renewable energy systems.

6.2 User Manual:

This user manual provides step-by-step instructions for using the Simulink model to change the irradiance and adjust the load reactive power for effective reactive power compensation.

1. Opening the Simulink Model:

- Launch MATLAB software on your computer.
- Navigate to the directory where the Simulink model file is located.
- Double-click on the Simulink model file (with the .slx extension) to open it in Simulink.

2. Understanding the Simulink Model:

- Familiarize yourself with the components and blocks in the Simulink model.
- The model consists of various blocks representing the solar PV array, STATCOM, capacitor bank, load, Statcom and other system components.

3. Changing Irradiance:

- Locate the Irradiance block in the Simulink model.
- Double-click on the Irradiance block to open its parameters dialog box.
- Adjust the irradiance value according to your desired simulation scenario.
- Click on the "OK" button to save the changes.

4. Adjusting Load Reactive Power:

- Locate the Load block in the Simulink model representing the reactive power load.
- Double-click on the Load block to open its parameters dialog box.
- Modify the reactive power value to simulate different load conditions or requirements.
- Click on the "OK" button to save the changes.

5. Running the Simulation:

- After adjusting the irradiance and load reactive power, click on the "Run" button in the Simulink toolbar to start the simulation.
- Monitor the simulation progress in the Simulink window.
- While running the simulation you can change the irradiance from the block.
- Once the simulation completes, review the results and analyze the performance of the reactive power compensation system, through the scope.

6. Analyzing Results:

- Utilize the simulation output data and visualization tools in Simulink to analyze the system's behavior and performance.
- Evaluate the effectiveness of reactive power compensation provided by the STATCOM and fixed capacitor bank under different irradiance and load conditions.

7. Saving and Closing the Simulink Model:

- After completing the simulation and analysis, save any changes made to the Simulink model.
- Click on the "File" menu in Simulink and select "Save" to save the model.
- You can save the plots from the scope .
- Close the Simulink model and MATLAB software when finished.

7 FUTURE WORK:

Flexibility to Grid Variability:

Design STATCOM systems with enhanced resilience to grid variability, including voltage fluctuations, frequency deviations, and harmonic distortions. Investigate robust control techniques and grid synchronization methods to ensure reliable operation under varying grid conditions.

Integration of multiple pv inverter:

We mainly designed the controller for one inverter. If there is multiple inverter in real life scenario, then values should be changed accordingly.

8 References :

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